



ACADEMIC CITY UNIVERSITY

DESIGN AND DEVELOPMENT OF VERIDIS, A SMART WASTE SEGREGATION SYSTEM WITH INCENTIVE REWARDS

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REWARDS**

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DECLARATION

We hereby declare that the research presented in this thesis is the original work of the undersigned students, carried out under the supervision of the named supervisor. The students and supervisor confirm that this thesis represents the outcome of the students' final year project work, submitted in partial fulfillment of the requirements for the Bachelor of Science (BSc) in Computer Science degree.



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ABSTRACT

The improper disposal of plastic and glass waste still constitutes a significant environmental challenge in the Ghanaian context, where recycling infrastructure is largely absent and behavioral incentives for waste segregation are minimal. In this study, we provide VERIDIS, an incentive-based smart reverse vending machine waste segregation system, based on the ESP32 microcontroller and the Flutter mobile application backed by Firebase, that automates the classification and sorting of plastic and glass bottles while rewarding users with simulated mobile money payments for their participation.

Bottles placed inside the machine are classified using sensor fusion combining a HuskyLens AI camera that performs primary visual classification based on a K-Nearest Neighbours algorithm and a digital LDR sensor that measures light transmission to distinguish transparent glass from opaque plastic. Each sensor votes independently and the outcome is determined by a majority voting algorithm. Depending on the result, servo motors driven by a PCA9685 PWM driver send the bottle into the correct bin via a sequence of mechanically coordinated gates and a bin divider. A 2.4-inch TFT display gives users real-time session feedback, while RGB LEDs provide signals of acceptance or rejection at each classification cycle.

Users start a session by scanning a QR code on the TFT screen, which triggers a deep link that opens the Flutter app. The ESP32 runs a REST API web server communicating with the app over a shared Wi-Fi network, handling session start and end requests, greeting users individually, and summarizing the session on the TFT. With user authentication, real-time bottle count tracking, transaction history, and a live leaderboard system that ranks users according to total contribution, adding a gamification element makes it easier to sustain users' engagement.

The success of the individual and partial integration of all major system components was verified with some tests. Hardware difficulties faced include the failure to detect the VL53L0X Time-of-Flight sensor on the I2C bus, and a pin constraint with the touch screen which relocated session end control to the mobile app. These limitations were noted as known for resolution in Phase 2. The results indicate that sensor fusion classification can be technically realized effectively on affordable embedded hardware and that mobile money incentives combined with gamification would offer a contextually appropriate motivator in the Ghanaian setting.

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Lastly, we would like to thank all the individuals who took the time, commitment, and effort to help make VERIDIS a reality. This project is proof of what we can accomplish when we align ourselves with a purpose.

DEDICATION

We dedicate this project to Almighty God for the wisdom, strength, and grace He has bestowed upon us throughout our academic journey.

We also dedicate this work to the collective effort, determination, and resilience that carried us through to the successful completion of this project.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Rapid population growth and urbanization across the world have resulted in a phenomenal growth in solid waste production. In 2050, annual waste production is projected to increase by 73% to 3.88 billion tonnes from 2020 levels (World Bank, 2022), threatening ecosystems and infrastructure around the globe. In the African context, recent projections suggest that the continent will produce approximately 516 million tonnes of solid waste every year in the coming years, with 90% of it destined for landfills due to poor collection and processing systems (Dugbazah et al., 2021).

This poor infrastructure leads to a rudimentary bottleneck, mainly the ability to properly filter waste from the source of its production. Plastic constitutes 13% of the waste stream (Organisation for Economic Co-operation and Development, 2022), and alternatives such as glass, when mixed with organic waste, become contaminated upon handling, which leads to their being contaminated, with their economic and recycling value falling drastically. This is very evident through the national waste crisis in Ghana, which, like these continental trends, has a direct counterpart in the national waste crisis there.

Ghana has been grappling with challenges within its waste management system, specifically with plastic pollution in their environment due to the country generating over 1.1 million tonnes of plastic waste per year, but with less than 10% of this recycled (National Plastic Action Partnership, 2021). Glass waste is not as prominent in policy discussions, it also builds up in dumpsites, where its slow decomposition and bulkiness further degrade the environment. The scope for improvement is huge, according to studies up to 97% of institutional waste in Ghana

is recyclable, but this potential remains almost entirely untapped due to segregation failures (Safo-Adu & Owusu-Adzorah, 2023). These materials together point to a gap in waste management and a need for innovative response strategies, especially in high-waste settings like university campuses. The combined information presented in this report indicates a significant issue in the way we currently handle waste management.

To close the gap in the way we currently handle waste management, a total paradigm shift is needed. The current practice of recycling at the end of the pipe must change to smart source segregation of materials. By using smart technology to enable the practice of waste segregation by making it more intuitive, rewarding and therefore, easier to track. This system makes it possible to transform the act of waste segregation from a chore into a habitual practice. The implementation of a technology-based gamification model with benefits will enable universities to leverage their existing digital activities and build sustainable waste management habits for their students.

1.2 Problem Statement

The waste management system in Ghana has a significant failure due to the fact waste is not segregated at the point of generation. This is a major system failure within the waste management system in Ghana, as studies indicate that institutional waste streams have the potential to provide an extremely high rate of resource recovery when compared to waste generated from other institutions. Studies have identified that institutional waste streams in Ghana contain more than 80% recyclable material (Dery, et al. 2018); however, all of this potential is lost at this point of the waste management chain.

The United Nations Development Programme (UNDP) recently reported that the plastics

recycling rate in Ghana continues to stagnate at about 9.5% (Saeed and Adodoadji-Dogbe 2025), which indicates a major implementation gap in national policies like the National Plastic Management Policy of 2020. The core problem is not a lack of intent from Ghanaians to segregate waste as 72% of people surveyed indicated a willingness to do so with proper infrastructure (Owusu-Ansah et al. 2022). The biggest barrier for Ghanaians is the lack of support and proper infrastructure needed to segregate waste.

Digital technologies have allowed for the global scaling of circular economy practices by providing benefits to participants. According to research conducted by Ranta et al. (2021), reward-based incentive programs can increase participation in recycling and segregation by 30–60%. However, none of the universities in Ghana have implemented these types of programs, which represents a significant missed opportunity given that students in Ghana are adopting mobile money services at a rapid pace and are enthusiastic about using digital technology.

This highlights an important gap because no single integrated technology-based platform exists across university campuses in Ghana that can motivate students to segregate waste at the disposal time. To bridge the gap, this study develops VERIDIS, a smart segregation station which leverages sensor-fusion technology along with artificial intelligence (AI) based material identification systems as well as a mobile money-based incentive. This is said to enhance both operational and motivational factors concerning sustainable waste management.

1.3 Objectives of the Study

1.3.1 General Objective

The VERIDIS system aims to achieve an increase in waste segregation, a target which has been grounded in research showing that reward-based incentive programs can increase recycling

participation by 30 to 60% (Ranta et al, 2021) and also improve overall environmental cleanliness on campus within a 6-month pilot deployment. By applying multi-sensor classification and mobile money incentives within a smart, integrated system, the system aims to achieve an accuracy of 90% or higher in sorting plastic and glass bottles, consistent with the greater than or equal to 90% accuracy demonstrated using the K210 AI chip in comparable waste classification deployments (Li & Grammenos, 2023), while instantly rewarding participants and encouraging students, as well as teaching and non-teaching staff, to participate.

The materials gathered are not wasted but recycled by partners and transformed into new products that can be used in everyday society, creating an eco-friendly environment and supporting a circular economy. Mobile money credit incentives can be accumulated and utilized for purchasing items on campus for users of the system. Using sensor fusion technology which is a combination of AI Vision, light-dependent resistors and infrared sensing, different waste types are detected and sorted into the proper bins, fully automating the entire process of recycling.

1.3.2 Specific Objectives of the Study

The specific objectives of the study are as follows:

1. To design and construct a reverse vending machine prototype that is able to accurately classify and sort plastic and glass bottles through a sensor fusion approach (HuskyLens AI camera, IR sensors, LDR sensors, and a Time of Flight sensor) with a target accuracy of 90% or greater.
2. To administer a mobile money-based reward system that presents users with instant rewards based on the total count of the segregated waste, to achieve an increase in proper waste segregation on campus.

3. To develop a mobile application with user authentication, transaction history, QR code session management, and a live leaderboard to encourage friendly competition and increase user engagement.
4. To initiate data collection protocols that track waste metrics (count, materials, and user participation) for partnership development with related organizations.
5. To validate system reliability through comprehensive testing, eventually achieving 95% or higher operational uptime during the 6-month pilot deployment, as a project-defined success criterion for prototype deployment readiness.

1.4 Scope of the Study

In this study, we will evaluate the effectiveness of the system, which consists of the correct classification and detection of the waste using sensor fusion with application of ESP32 microcontroller, the accurate presentation of a real-time leaderboard, efficient user management and the proper allocation of incentives with respect to total count of the waste recycled. More specifically, the system will focus on the collection and classification of plastic and glass bottles, which are two of the most common and problematic types of waste on campuses. Our system brings together hardware (an ESP32-based reverse vending machine with a HuskyLens AI camera, IR sensors, and LDR sensors) and software (mobile app and mobile money reward system) into a working prototype. The system is intended for use by university students and staff. The research is carried out not for mass production or long-term use, but for prototype development and would presently not be employed to enable the disposal of any other types of waste.

1.5 Significance of the Study

This study exists to raise solutions to issues that are related to littering and the improper disposal of waste, mainly at universities in Ghana. Employing a unique method of combining the HuskyLens AI camera, infrared sensors and light detection sensors, VERIDIS will demonstrate how technology has the potential to change recycling behaviors.

The significance of this study lies in the following key contributions:

1. **Technical Innovation:** By integrating sensor fusion into a low-budget, cost-effective, and scalable ESP32-based system, the system can assist in automating waste classification in a resource-poor or low-resourced environment. Research has shown that sole reliance on single-sensor RGB imagery leads to systematic misclassifications between glass and differently colored plastics (Ahmad et al, 2024).
2. **Behavioral Impact:** By providing instant rewards in the form of mobile money for users in relation to the waste segregated, VERIDIS creates tangible incentives that resolve the motivation gap in current recycling efforts.
3. **Practical Implementation:** VERIDIS serves as a replicable blueprint for campus sustainability initiatives, exhibiting how a gravity-based design and smart sensor technology can create maintainable, power-efficient waste segregation systems suitable for the Ghanaian context.

1.6 Limitations of the Study

While VERIDIS is an innovative approach towards campus waste management, it is subject to many constraints that could affect its scope, performance, and impact:

- **Technical Constraints:** The sensor fusion system may encounter classification

challenges with soiled, distorted, damaged, or unusually shaped bottles that deviate from the trained patterns, which will potentially reduce the target 90% accuracy rate. Environmental factors such as inconsistent lighting, dust, and humidity could also cause sensitivity problems for the optical sensor, resulting in the need for ongoing and frequent testing of each sensor.

- **Resource Constraints:** The restrictions placed on project funding limits the types of sensors and construction materials available for use in the sensor fusion system. In addition, the type of test facilities available to conduct endurance tests to demonstrate the durability and long-term reliability of the prototypes with continuous use are also severely limited. The project deadline (26-30 weeks) also restricts the time available to conduct extensive long-term and iterative testing, as well performing validation on multiple campuses.
- **Implementation Scope:** The system has been set up to only detect plastic and glass bottles and no other recycling materials such as paper, aluminum cans, and food packages. The use of mobile money depends on the availability of an API and the transaction fees associated with the API that will determine how much a user can earn by using the VERIDIS system. Although the gravity-fed mechanical design is power efficient, it may jam if not calibrated to detect non-standard-shaped containers.
- **Operational Constraints:** This study focuses on prototype validation rather than large-scale deployment, which may limit the assessment of long-term maintenance requirements and operational costs. User adoption could be influenced by the usability and learnability of the mobile app as well as the perceived value of rewards in relation to the effort required.

1.7 Organization of the Study

The study is organized into five (5) chapters that allows for an organized approach towards the research and development of VERIDIS: a smart waste segregation system with mobile money incentives.

Chapter one is the baseline for developing a sophisticated waste segregation incentive-based system on University Campuses across Ghana.

Chapter two comprehensively reviews and discusses existing research and the designs of existing RVM and incentive-based waste systems, AI in waste classification, sensor fusion, and mobile money payment integration. It identifies gaps in current systems and establishes proposed solutions.

Chapter three details the system architecture, hardware design, software development, mechanical design, and implementation approach. It includes all system diagrams, component specifications, and testing methodologies.

Chapter four discusses the development and testing of VERIDIS, detailing the performance of hardware components like the TFT display, HuskyLens AI camera, infrared sensor, LDR sensor, RGB LEDs, and Wi-Fi connectivity. It explains the classification logic and majority voting system, as well as features of the mobile app, including session management (QR), Firebase integration, mobile money rewards simulation, leaderboard, and transaction history. Lastly, it presents survey findings on student awareness, willingness to use the system, and mobile money familiarity.

Chapter five concludes the study and makes some conclusions on the results presented in chapter four. It demonstrates the technical feasibility of the VERIDIS architecture and

hardware and mobile application layers integrations. It presents the drawbacks of the development phase. It ends with suggestions on phase two development, and addressing the problems in the hardware that the system encountered.

CHAPTER TWO

LITERATURE REVIEW

2.1 Overview of the Project

This chapter presents a deep dive into the primary concepts and technology that shape the design of the proposed Reverse Vending Machine (RVM) System. It provides information regarding previous studies performed in regard to automated waste management, incentive-based recycling Systems, AI -based waste classification systems, sensor-assisted object identification and reward systems. The intent of the chapter is to demonstrate the technical and theoretical basis for these design choices and to highlight the areas where the project will provide improvement.

RVMs are automated systems that take in waste materials and, in return, dispense rewards in the form of tokens or discounts to depositors in order to amplify environmental cleanliness and recycling (letsrecycle, 2025). These systems employ artificial intelligence in analyzing and classifying deposited waste materials and then share appropriate rewards based on the outcome of the analysis(Nasir & Al-Talib, 2023) . Modern RVMs incorporate computer vision and object-recognition algorithms to verify material type, improve sorting accuracy, and reduce contamination in recycling streams.

Rewarding-based waste systems operate under the understanding that rewards encourage recycling behaviors (Ranta et al, 2021). In this context, deposit refund systems serve as a pertinent example, illustrating a model in which an initial deposit is returned upon the item's return to the retail establishment. The literature presents a novel approach to incentivization through an innovative blockchain-based tokenization initiative. This method involves the digital creation of value units on the blockchain, facilitating transparent and tamper-resistant

tracking of rewards (ABC Research, 2022; RecycleGO, 2021). While systems such as RecycleGO have demonstrated practical efficacy, VERIDIS adopts a distinctive approach tailored to the Ghanaian context. This approach involves the distribution of mobile liquidity through platforms such as MTN Momo, serving as a strategic vehicle for implementation. This approach utilizes existing financial infrastructure without putting end users on the blockchain. These developments are further compounded with the proposed project which proposes to marry machine-learning classifying models, sensor assisted object detection for placement precision and mobile money secured reward distribution.

The use of these combined technologies is not just to improving recycling, but also aiming to help overcome challenges of existing systems such as inaccurate identification of waste, manual sorting and low user participation.

2.2 Related Works

2.2.1 Plastic and Glass waste Management

Environmental issues are affecting every single country due to their plastic and glass waste. Over the last 40 years, various reports have established that the amount of plastic produced consistently rose since 1980, resulting in an annual production of 360 million metric tons per year, and is projected, based on existing estimates, to reach a global plastic production of 2 times the current levels (around 360 million metric tons per annum) and is predicted to happen by 2040 (Statista Research Department, 2025). Currently, worldwide glass production is around 130 million tons yearly (Baek et al., 2025). Only 21% of the global glass production is recycled (Baek et al., 2025). While international organizations such as the UN have made efforts in this regard, the production-recycling gap is continually widening, leading to a number of sustainability debates and increased environmental impact.

In Africa, substantial amounts of waste are generated, with collection rates averaging at approximately 55%. Only 4% of the total amount of plastic waste is being recycled, despite the fact that about 70-80% of it could be recycled (Nepad, 2021). Many countries located along the coast of West Africa have problems with the management of plastic waste and this figure shows that about 80% of this is being disposed of improperly. In Ghana, it is estimated that approximately 840,000 tons of plastic waste are produced every year, with only a small portion being properly recycled (Daily Statesman, 2024). Research indicates that plastic and glass waste accounts for 21.95% of all solid waste generated in this country (Agyei, 2020). The large quantities of single-use plastics being purchased and irresponsibly discarded and the lack of an adequate waste management system are major factors in this statistic.

In Ghana, finances are also highlighted as a major developmental constraint for waste management (Mesti, 2020). Ghana also acknowledges weak strategic, institutional, and organizational structures that have been perpetuated by limited skills essential to waste management and revenue generation for the cause (Nepad, 2021). Poor management of plastic waste in Ghana has resulted in significant environmental, health, and economic challenges. Studies undertaken in Accra show that large quantities of plastic debris are disposed of indiscriminately in landfills, open spaces, and especially in gutters and drains. This accumulation contributes to land and water pollution and increases the risk of flooding, particularly in urban areas such as Accra and coastal cities like Cape Coast (Squire, 2016).

Glass shards as well pose a threat to aquatic life, local communities working water bodies and people visiting beaches.

Though Ghana has enacted an Environmental Sanitation Policy, local government regulations and other laws aimed at improving waste management, enforcement has been slow and hence ineffective (UN, n.d). Universities are playing a key role in developing new innovative solutions for better waste management throughout Ghana, specifically around the issue of plastic pollution. Ghana generates an estimated 1.1 million metric tons of plastic waste each year, with only 5% being recycled. Thus, many universities have proactively created a waste management system that will serve as an example for the entire country (Debrah, 2025). One project that started at KNUST in 2017 was the implementation of waste segregation and collection through the deployment of 332 color-coded waste bins (yellow for plastic, green/black for everything else) located throughout the KNUST campus. Collected plastic waste is sent to an on-campus recycling factory that produces crushed and pelletized (plastic) products. This facility is designed to process two tonnes a day of pelletized plastic and five tonnes a day of crushed plastic through its palletization and crushing equipment respectively.

In 2025, however, a subsequent study revealed several major obstacles preventing optimal operations of the manufacturing facility. Less than half of campus residents (43.9%) knew of the waste segregating system, while about 75.5% reported having personally encountered the designated bins, and only about 56.5% had understood the various colour-coded bins denoting the appropriate category of recyclable material. This study found that only 46.1% of survey respondents practice waste separation, while their average percentage of identified plastic placed correctly in the yellow recycling bin reached an average percentage of just over 69.36%, indicating that less than three-quarters of all participants had achieved the desired complete separation. Key barriers highlighted included low publicity, insufficient bins, and inadequate monitoring.

These realities illustrate the necessity of novel and adjustable waste management techniques. Techniques, such as reverse vending machines and reward-oriented systems, are highly effective in encouraging increased recycling rates by altering how people conduct their recycling. Therefore, universities/campuses provide a strategic place to assess these strategies since they allow for controlled environments to evaluate, change, and expand their recycling habits into the larger community.

2.2.2 Reverse Vending Machine

Reverse vending machines (RVMs) have been developed over recent years as an innovative approach to solving issues related to managing solid waste. An RVM is a type of machine in which users can place their used or empty rubbish (bottles and cans, etc.) into the machine at which point a scanner will identify the item and sort it based upon the material type (glass vs plastic etc.). After sorting, users are issued an incentive (vouchers/tokens) for returning their items to RVMs to encourage recycling. The ease and incentives associated with this type of recycling will hopefully motivate users to recycle their used glass and plastic products (letsrecycle, 2025).

The use of reverse vending machines around the world is now being done primarily in developed countries, especially within Europe and Asia. Germany has arguably seen the most success globally due to their deposit return system achieving a 98% recycling rate for beverage containers. The system is supported by thousands of reverse vending machines throughout the country (Reloopplatform, 2024), (TOMRA, n.d). Since 1996, Norway has successfully implemented its Infinitum system, which has resulted in an astounding 97% recycled rate for plastic bottles. This system utilizes approximately 3,500 reverse vending machines, which have collected approximately 93%, by volume, of all returned containers (Reloopplatform, 2024).

In Asia, Japan operates between 600 and 1,300 reverse vending machines throughout the country. The system collects around 9,740 tons of PET (polyethylene terephthalate) bottles annually through 820-1,001 installed machines. The Japanese consumers receive “2 recycle points” (0.2 yen) per plastic bottle, which can be used in stores for purchases (Tang, 2011). Hong Kong’s pilot program serves as one of Asia’s most comprehensive RVM initiatives. The government-funded project operates 120 RVMs across strategic locations, successfully collecting over 60 million plastic bottles in just one year (Alba, 2025).

Unfortunately, the situation in Africa is vastly different. The development and use of Reverse Vending Machines (RVMs) on the continent have many barriers stemming from structural, economic and social factors which contribute to the difference in how developed markets operate versus how Africa operates (Litoclean, 2024). A number of African regions have unreliable electricity grids, which creates a very serious problem regarding the deployment of RVMs (Plu Team, 2024). Some newer machines, such as the RVM S01 by Eco Barter in Nigeria, have been developed with deep-cycle batteries to allow for up to 48 hours of backup power; however, the lack of consistent electricity supply causes significant problems associated with deploying RVM technology (Plu Team, 2024). Modern RVM technology is complex and requires substantial amounts of capital investment, but many municipalities and businesses in developing regions do not have the financial ability to make such investments, therefore creating issues regarding RVM deployment in many parts of Africa. In addition, local governments in Ghana either have limited resources or do not generate enough revenue from their current operations to make RVM investments a viable option without it being subsidized by another organization or agency. As a result, it is important to find locally adapted designs for RVMs rather than directly copying the RVMs from Asia and Europe.

Reverse vending machines have highly relevant and transformative tools for university campuses, addressing multiple institutional goals while providing educational, environmental, and social benefits to the campus community. According to reports, universities that implemented the RVM(s) have seen a significant number of educational advantages. The RVM system provides students with an opportunity to learn through active participation and enhances their understanding of both environmental and circular economy-related concepts. The RVM installation at the University of Glasgow (Gla 2018), was created specifically for the purpose of increasing sustainability on campus. The Durban University of Technology (DUT) launched six RVMs across its campuses in September 2024, positioned at strategic locations including libraries and student canteens. These machines reward users through the Imagined Earth app with digital credits redeemable for airtime, Easy Equities investments, and Shell V+ points (Durban University of Technology, 2024).

2.2.3 Incentive-Based Waste Systems

Systems that are incentive-based to encourage waste management utilize the principles of Behavioral Science to encourage individual participation in recycling through tangible rewards for recycling behavior. Individuals will participate more in recycling activities when they receive a reward (i.e. token, cash, discount) than if they are not given an external reward. The external rewards produce immediate, tangible benefits to the individual's quality of life. The relationship between the rewards system and recycling behaviors are not so straightforward as it might appear because there are many, more complex, psychological dynamics to understand regarding an individual's motivation to recycle (ie. internal (intrinsic) and external (extrinsic) motivational dynamics). Intrinsic motivation describes the internal drive to take part in positive environmental behavior because it aligns with personal values or feels morally right (Sustainability Directory, 2025). In contrast, extrinsic motivation is driven through external

rewards, consequences, or social pressures. When recycling is extrinsically motivated, people participate to obtain monetary rewards, avoid penalties, or gain social approval.

Various global models demonstrate the efficacy of the approach. Two pioneering firms, RecycleGo and EcoATM, illustrate different approaches to incentivizing recycling behavior through technology-driven reward systems, demonstrating significant success in boosting participation. EcoATM enables customers to drop in their old devices, such as old phones, MP3 players, and tablet, to the kiosk, where device specifications are evaluated and payment offered in a matter of minutes. EcoATM has established itself as a scalable e-waste collection platform, showing that immediate compensation with a kiosk can greatly reduce the threshold to responsible device disposal (EcoATM, 2023).

Moreover, RecycleGO provides a recycling technology platform that leverages blockchain technology to help users monitor, validate, and reward sustainable waste management. The platform uses Hyperledger Fabric Blockchain Technology to automate reward and smart contract mechanics, as well. RecycleGO automatically creates and provides tokens based on the recycling activity of disposable waste products to a User's digital wallet automatically (RecycleGO, 2021). Both platforms in particular are proving that enabling reward-based programs on recycling to address the four main behavioral key drivers (convenience, immediate feedback, fair compensation, and a clear environmental impact) leads to very large user engagement. Tokenization and digital forms of digital payment are cutting-edge technologies which have fundamentally transformed the recycling reward ecosystem in ways it often has not in the past. They allow waste management systems to be more efficient, accessible and scalable than ever before. Digital payment integration has completely transformed the recycling incentive system through removing the barriers that paper vouchers, cash, and manual processing presented, keeping users from taking part in these programs in the past. In addition, many of today's more modern Reverse Vending Machines (RVMs) are

also now integrated with digital wallets, mobile apps, and contactless payments making it possible for users to process their reward payments in one seamless transaction (TOMRA, 2022). A perfect example of this transformation through technology would be TOMRA's Digital Payout API that allows customers to receive their deposit returns directly to their loyalty program app/store apps or digital wallets, rather than receiving them via physical paper vouchers. To date, there have been approximately 30,000 digital transactions generated through this integration in Belgium, and the trend is growing.

The rapid expansion of the mobile money platforms within the Ghanaian landscape with companies like MTN's Mobile Money and Vodafone Cash represent an opportunity to integrate recycling incentive programs directly into established forms of payment accepted within financial institutions in Ghana. The growth of volume of transactions processed within Ghana's mobile money sector has been monumental. In 2024 alone, a total of 3.02 trillion Ghana Cedis has been processed, with a 57.90% increase in volume compared to last year. Therefore, if we integrate this system within the RVMs, it would allow for ease of participation and availability for individuals residing within geographical locations lacking sufficient access to banking infrastructure.

2.2.4 AI in Waste Classification

Waste classification is the process of identifying and categorizing waste materials based on their composition. This process is often done in order to properly dispose of and recycle the waste in a responsible manner. There are various ways of classifying waste, which include visual inspection, chemical analysis, and the use of machine learning algorithms or artificial intelligence (Nasir & Al-Talib, 2023). Artificial intelligence, particularly methods and techniques from computer vision, has been proved to be an effective tool for automating waste classification. AI is deployed to automate and accelerate the sorting process, a critical response

to today's rapidly increasing volumes of waste generation (Kaya et al., 2023). In order to achieve this, there are tools or architectures under AI that are primarily used to complete the task of classifying waste accurately and fast.

Convolutional Neural Networks (CNNs), a class of deep neural networks, are the most widely used architecture under AI for image-based classification tasks, and this includes classification of waste under the various waste materials (for example, plastic, glass, metal, and paper). CNN is a deep learning algorithm that processes visual data by applying filters that convolve across an input to detect patterns such as edges and textures, using activation functions and pooling layers to make classifications (Bhalerao & Ingle, 2021). CNNs are considered the “gold standard” for object and waste classification solely due to their exceptional ability to automatically learn and extract necessary features from visual data. Compared to traditional machine learning (ML), where feature extraction (color, texture, and shape) required manual entry, CNNs learn all the necessary features directly and automatically from the raw visual data (Mao et al., 2020). In one study, a CNN model was used to classify waste into seven categories, which are cardboard, glass, metal, organic, paper, plastic, and trash (Poudel et al., 2023).

While accurate, training a CNN from scratch needs substantial computational resources and time. Pre-trained models, on the other hand, have already learned the generalized features from large datasets, and this allows for generally better accuracies, especially on smaller datasets. This study employs the HuskyLens AI vision sensor, an AI vision sensor with built-in machine learning algorithms. It processes visual information for tasks such as facial recognition, object tracking, line tracking, color recognition, gesture recognition, and in our case, object recognition. The HuskyLens uses a pre-trained model to perform its tasks. It makes use of a built-in Kendryte K210 AI chip for visual information processing. This chip has a dedicated KPU (Neural Network Processor), which is a hardware accelerator for CNNs, capable of up to

0.8 TOPS (Tera Operations Per Second) of computing power. The K210 chip can achieve accuracies of above 90%, and the device only consumes about 0.8 W at inference, making it power efficient and even capable of running on solar power. The chip is also faster at booting up, taking only 5 seconds, and immediately starts detection when switched on (Li & Grammenos, 2023). Some K210 chips use a network called tinyYOLOv2 (You Only Look Once version 2), which is a lightweight, accurate, real-time model based on the YOLO architecture for object detection. This tiny network is easily compatible with resource-constrained embedded devices while maintaining accuracy (Saha et al., 2025).

The integration of AI in waste classification can lead to significant performance improvements if done right; for instance, Li & Grammenos (2023) observed accuracies of 90% or more using the K210 over different trials in waste classification (Li & Grammenos, 2023). Despite these performance improvements with image-based classification, several limitations remain. Most AI-based classification models make use of data that lack diversity, do not reflect real-world conditions, and/or are unbalanced, which results in these models' failure to perform effectively when applied to different urban systems (Ahmad et al., 2024). Ahmad et al. (2024) also observed that even with high accuracies, misclassifications occur between different colored and clear glass and plastics, which highlights the limitation of sole reliance on RGB imagery. There are also shortcomings, such as accurate detection of range and depth, that are vital challenges for AI-based or camera-based classification (Xie et al., 2020).

While AI vision like the HuskyLens AI camera offers high accuracy, this study acknowledges its limitations in real-world conditions, particularly with material analysis and varying lighting. This goes to highlight the need for a multi-sensor or sensor fusion approach to improve overall robustness.

2.2.5 Sensor Fusion in Waste Classification

In this study, a sensor fusion approach is employed to overcome the limitations of a solely camera-based waste classification. Sensor fusion can be understood as the process of combining data or signals acquired from multiple sensors to create a more reliable, robust, and accurate representation of the environment, thereby reducing uncertainty and improving performance, resolution, and confidence (Elmenreich, 2002).

To achieve this fusion of sensors, a notable sensor for object detection and material analysis or identification is the infrared sensor (IR sensor). Infrared sensors are sensors that operate by emitting and detecting infrared light, which is invisible to the human eye but can be emitted by objects, and measuring the intensity reflected from the surface of the material. Different materials show different reflectance patterns. This sensor consists of an emitter, which is simply an IR LED, and a receiver, which is simply an IR photodiode that is sensitive to IR light of the same wavelength as the emitted IR light (Goud et al., 2023). These sensors serve two separate but crucial roles in waste systems, which are object detection and material analysis for sorting. In Goud et al. (2023), IR sensors were used to detect the presence of waste, and this triggers an automated sorting mechanism that directs materials into their respective containers. Basically, whenever the system enters rest mode and an item is dropped, the IR sensors wake the entire system up to start processing. Another study made mention of IR sensors for material analysis or material type analysis based on material reflectance for waste classification (Argirusis et al., 2025). In order to further improve the accuracy of the system when it comes to waste classification and automate the entire waste segregation process efficiently, VERIDIS employs the use of IR sensors for object detection for system activation and material analysis for sorting. Even though regular IR sensors have good waste classification accuracy, they encounter essential limitations. Such sensors cannot be used for dark-colored or black plastics (Ruj et al., 2015). This imposes a limitation due to the large

number of dark plastic bottles found in the waste streams in Ghana. IR also faces challenges with multi-layered or composite materials, which are common in modern waste streams (Araujo-Andrade et al., 2021).

Another notable sensor that will be employed in this study is the Light Dependent Resistor (LDR) sensor. An LDR sensor, also known as a photoresistor, is a sensor that operates by converting light energy into electrical resistance variations, where higher light intensity reduces resistance and lower light intensity increases resistance (Annas et al., 2022). LDR sensors can be used for object detection, object distinction, and bin fullness in waste systems. This study utilizes the use of this sensor in object distinction, to be more specific, opaque and transparent material distinction. LDR sensors can be used to tell the difference between opaque objects and transparent objects based on their transmittance and reflectance. Opaque objects, such as dark-colored or colored plastic, will scatter most of the light thrown at them and only reflect very little of it back to the LDR sensor, meaning the resistance is high and the reading will be low. On the other hand, transparent objects, such as clear glass or clear plastic, will allow a significant amount of light to pass through them and only reflect more light back to the sensor, meaning the resistance is low and the reading will be high. The use of the LDR sensor offsets the limitations of IR sensors (Annas et al., 2022). Though the LDR sensor offers its benefits, it also has some key limitations, which include sensitivity to temperature and slow response times (Annas et al., 2022).

While camera-based classification offers rich semantic information like color, texture, and shape, which are all essential for waste classification. They are still sensitive, if not highly sensitive, to environmental conditions (lighting, occlusions, and shadows), and lack accurate material and reflectance information. The proposed sensor fusion approach for VERISDIS, with the AI vision, IR, and LDR sensors, complements each sensor's strengths, which in turn significantly enhances the classification accuracy and performance and the robustness in real-

world operating conditions. This approach makes VERIDIS best suited for the campus environment.

2.2.6 Mobile and Web Applications for RVM Interaction

The modern recycling system is increasingly dependent on the digital interfaces that connect users to the process. Mobile and web applications transform the act of waste disposal from an anonymous transaction into a tracked, engaging, and incentivized activity. Recent research by Adaji et al. (2024) confirms that a well-designed digital layer is not merely an add-on but a core component to ensure participant adoption and long-term system viability.

Waste management is a topic in which there is a substantial amount of research supporting the idea of incorporating the Internet and electronic tracking systems as part of how we manage waste. Jagannathan et al. (2022) completed a study that created a digital audit trail for municipal workers to scan QR codes located at dust bins, using a mobile device application that allowed the municipal worker to monitor household waste segregation compliance. The study showed that the creation of a unique identifier can offer a valuable method of accurately recording the origin of waste materials. It can also be applied to RVM interaction where a QR code or user ID can authenticate the user for accurate attribution of recycled materials and the personalization of the process itself. It is precisely this personalization that allows self-monitoring of behavior, which Adaji et al. (2024) determined was the most widely used persuasive strategy in effective sustainability apps. User experience (UX) design of these applications has also unique challenges apart from basic functionality. As the concepts above (note), user onboarding and rewards clarity are key aspects to adoption. Users may be disappointed by a confusing initial setup, and the lack of reward details could be a deterrent. Adaji et al.'s review of user reviews (2024) indicates that positive user experiences are described as “easy,” “great,” “brilliant,” while negative reviews more often mention usability

problems and underscore the importance of intuitive design.

In addition, the processing of user data is another huge added layer of complexity. Kretschmer et al.'s survey (2021) of the General Data Protection Regulation shows users face nebulous data usage policies and vague privacy policies which can turn off users and erode trust. For platforms designed to keep user credentials and offer incentives for users to use those credentials, ensure transparency and data security from the beginning to create and maintain user trust. Writing in the literature suggests implementing gamification to minimize low participation levels. “Methods for increasing student engagement involve maintaining leaderboards that promote friendly competition and rewarding recycling streaks,” the authors wrote. Exploiting techniques informed by behavioral psychology, they attempt to make recycling entertaining and socially desirable. This aligns with Adaji et al. (2024), indicating that apps using a variety of persuasive strategies are more likely to lead to behavioral change. Adding these elements allows a mobile app to turn recycling from a chore into an enjoyable and habitual activity. While gamification offers advantages, when used as a standalone strategy it is not a one-size-fits-all solution.

Recent results by Venturi et al. (2025), represents an alternative critical understanding of gamification, wherein the potential of gamification with possible engagement enhancement for some remains incomplete and provides the basis for discussion. In their experimental study, students found that a gamified recycling app did not increase knowledge or correct recycling behavior more than a non-gamified recycling app with clear and on-demand information. Specifically, game mechanics are seen as a “waste of time” and a hindrance to learning by many users, and this resistance is particularly high for older groups. A digital recycling tool is valuable only at scale if and when it can be accessed. With the VERIDIS system, that translates to a user-onboarding that aims to recruit a younger user base paired with immediate reward explanation.

Currently, the best digital strategy for incentivizing sustainable behavior has is a system built upon basic utility and a coherent experience for the customer as a whole, where gamification is built in to supplement, not compete with, the basic recycling process.

2.2.7 Gaps in Existing Systems

Recent literature reviews of waste management have exposed some shortcomings of existing systems in the African context. The proposed system is an empirically validated, direct, integrated response to these specific gaps.

The lack of effectiveness in awareness-based campaigns and manual segregation is one key issue. It has been reported that 67.4% of a national sample in Ghana were aware of waste segregation but the awareness did not facilitate implementation (Owusu-Ansah et al., 2022). Another recent evaluation of KNUST's bin-centred system in 2025 found that it was largely ineffective (Sackey et al., 2025), with only 43.5% of the campus community understanding bin color codes. VERIDIS is specifically set to address this gap to automate the sorting and sorting-out process entirely. Using artificial intelligence vision and sensor fusion, it delivers highly accurate classification without the need for user knowledge, thus eliminating the high error rate of the systems that rely on manual methods. These behaviors are a major bottleneck of the massive recycling eco-system.

Although SESA Recycling and similar companies have the infrastructure capacity to process the plastics they collect, users still use conventional separation methods, which regularly result in the dispensation of contaminated materials. Furthermore, the 30.6% contamination rate (Sackey et al., 2025), recorded on campus systems, directly influences recyclers' costs and efficiency, which is a core supply chain constraint at a local level. VERIDIS enables the contamination problem facing the full recycling value chain to be avoided by guaranteeing clean material streams through automated source segregation. In addition, the financial price

of the waste is divorced from the potential value of recyclables. The scale of such economic inefficiency is almost mind-boggling. Dery et al. (2018) examined this issue and found that a Ghanaian university was using more than GHC 193,000/year for disposal and discarded recyclable paper worth GHC 7,830/year for completing the process. That's a huge financial drain for institutions that enter into that waste because they pay a price for losing precious resources. VERIDIS fills that void with instant mobile money rewards and a new kind of economy. Its true cost is turned to an economic potential stream. The second feature is the efficiency of its cash. In this manner it closes the resource-value loop, since it dedicates some of the savings from the lower level of contamination and more cost-effective disposal to incentives for users and the incentives associated with it. Also, the need for smarter, data-driven management is a void we all need. At present, the methods of evaluation are reactive in nature, primarily using manual sampling methods which potentially do not provide the maximum opportunity for real-time assessments. Future researchers have called for future research to specifically investigate real-time monitoring systems for assessing waste collection (Akomea-Frimpong et al., 2023). As an IoT integrated solution, VERIDIS is in itself built for addressing this requirement. On each one, live participation information, materials flow and system performance are generated, to enable the waste to be proactive and optimized.

The third area that current systems fail to access is the unexploited recycling potential. Waste characterization studies show consistently high rates of recyclability among Ghanaian institutions, which have been reported to reach up to 97% (Safo-Adu & Owusu-Adzorah, 2023). But implementing automated sorting at the source is still rare. VERIDIS serves as the most important technological platform to exploit this promise. Through the AI classification to make sure that products are clean, source-labeled and source-segregated, it contributes to the immediate value addition of recyclables, and hence responds to the demand for efficient "sorting algorithms" in the developing countries.

Finally, the proposed system does not simply identify problems; instead, it serves as an integrated solution, which aims for the primary technological, behavioural and economic shortfalls that currently exist in the country's waste management systems in Ghana and its for example.

CHAPTER THREE

METHODOLOGY

3.1 Hardware System Implementation

3.1.1 System Requirements

Hardware Functional Requirements:

- Ability to detect the presence of an item at the machine entrance using a proximity sensor.
- Ability to capture and analyze item images using the HuskyLens AI camera for material classification.
- Ability to identify materials (plastic vs. glass) using LDR and IR sensor output.
- Ability to count total bottles to provide a reward.
- Ability to operate the sorting mechanism via a servo motor-controlled gate, based on classification results.

Hardware Non-Functional Requirements:

- The bottle sorting classification process should take between 3 and 5 seconds.
- The system should be able to continuously function for 8 or more hours without performance deterioration.
- The machine should recover from power fluctuations without losing any data.
- The display interface should have consistent colors and icons.
- The machine should have a visually appealing design appropriate for public spaces.

3.1.2 ESP32 Microcontroller

On VERIDIS this is the main processing unit where we use the ESP32 microcontroller to orchestrate all of the hardware components, interpret data coming from sensors, and run a decision-making algorithm. The ESP32 was selected because of its low power consumption,

integral Wi-Fi and Bluetooth capabilities, quick processing speed, and ability to work with several sensors at once. The ESP32 will govern communication between the HuskyLens AI camera, IR sensors, LDR sensor, load-cell amp, servo motor, and TFT display inside this project. The ESP32 will analyze the data received from the sensors to conclude if the inserted item was valid and calculate the reward associated with the item and to execute mechanical movements to open/close the disposal gate. The tiny size, adaptable capabilities, and fast processing make the ESP32 an excellent microcontroller option for any embedded system, including the reverse vending machine.

3.1.3 HuskyLens AI Camera

In the reverse vending machine, the use of a HuskyLens AI camera will allow for object classification and identification. The HuskyLens has built-in machine learning capabilities for recognising recyclable objects from its dataset, so it can be used to identify items in real time. Once a user drops in an object, the HuskyLens will take an image of the object and compare it against what it has been trained on to see if it matches up as either a bottle made from plastic or glass. The ESP32 board interfaces with the HuskyLens using either I2C or Serial, to provide back the classification result. As established in section 2.2.4, the onboard Kendryte K210 AI chip processes visual classification tasks independently of the microcontroller, achieving accuracies of 90% or above in waste classification trials(Li & Grammenos, 2023), which reduces processing load on the ESP32 and supports the system's 3 to 5 second classification target. Because of this, the HuskyLens is an important part of the decision-making process of the reverse vending machine.

3.1.4 Infrared Sensors

Infrared (IR) sensors are electronic devices that detect objects or measure changes in their surroundings by emitting and receiving infrared light. In the RVM, two distinct types of IR sensors will be incorporated, each serving a specific and complementary function within the waste detection and validation process.

First, we will be putting in a simple IR break-beam sensor at the point of entry as a “first item” sensor. When a user inserts a can or bottle into the unit, it interrupts an infrared beam from the sensor creating a trigger event for a processing sequence indicating that an object is present. The HuskyLens AI camera and other types of systems will only power on when there is an object which saves power while also allowing increased functionality. The VL53L0X Time-of-Flight (ToF) sensor will perform material analysis inside the closed classifier chamber, whereas VL53L0X is not considered a reflectance sensor, it provides highly precise pulses of infrared light and measures the time it takes to return. As mentioned in section 2.2.5, ToF based infrared (IR) sensors take into consideration both return time, as well as the characteristics (amplitude) of the returning signal, which vary greatly based on the material that the object is made of (for example, plastic will return a different signal compared with glass compared with metal) (Argirusis et al., 2025). This method uses several types of sensors to provide information to verify classifications. In addition to visual classification information provided by the AI camera, these sensor measurements provide us a reliable representation of what materials are in a beneficiation waste pile using both fusion and multiple sensor approaches to accurately classify the waste and exclude items that should not have been included. The two IR sensors work together because they provide complementary data to get a complete and reliable classification. The break-beam sensor provides the means for system trigger, and the VL53L0X ToF sensor provides a critical layer of material verification. If integrated properly, the end product is a fully automated waste classification process that is both dependable and accurate.

3.1.5 Light-Dependent Resistor

The system will use a Light Dependent Resistors (LDR) as part of the process for identifying items deposited into the Classification Chamber. When an item is placed in the Classification Chamber, a controlled light source (LED) will illuminate it and the LDR will measure the amount of light transmitted or reflected by the item. Material transparency (glass) makes it more effective in transmitting light through, leading to an increased signal at the LDR. Plastic materials (or opaque materials) block or absorb more light, making for a lower signal. This information is then brought in conjunction with information from the IR Time of Flight sensor in addition to images taken by the HuskyLens artificially intelligent camera, which helps improve classification accuracy. By differentiating between opaque and clear materials, the LDR plays a critical role in making sure that items are returned to the correct bin to get there in such a way as to make sure that recyclable materials are sorted out from non-recyclables.

3.1.6 Light Source (LED)

An LED light source will provide the illumination in the detection chamber of the reverse vending machine. The HuskyLens AI camera needs enough illumination so that it can recognize items, regardless of the level of outside ambient light. The LED is activated automatically by the ESP32 when the break-beam IR sensor detects an item at the point of entry. Due to its energy-efficient design, long life, and steady output, the LED forms an important part in correctly classifying images. In addition, including this component allows the system to reliably perform in a myriad of different environment types, including low-light conditions. The light source also serves as the environment for LDR work by determining the quantity of light passing through the object.

3.1.7 Proximity Sensor

A proximity sensor is a non-contact system that senses the presence of nearby objects by emitting and receiving electrical signals, such as infrared or electromagnetic waves. The following project uses the proximity sensor to monitor the fill level of waste bins in the Reverse Vending Machine. The sensor will then detect when recyclable items accumulate and reach a predefined height and send a notification when the material is close. This feature means that the system is able to alert the administrator automatically when either bin is nearing full capacity. The proximity sensor ensures optimal work through timely notifications, preventing overflow, and promotes smooth machine operation with effective maintenance scheduling. Due to its non-contact nature, it is reliable and thus suitable for continuous monitoring in waste-filled environments.

3.1.8 TFT Display

The TFT display integrates with the recycling process, and will provide real-time feedback to the user. It will show system messages such as, “Insert Bottle”, “Processing”, “Accepted”, “Rejected,” and “Reward Credited.” This again increases usability, and ensures that people understand every interaction. ESP32 controls the display and the display is responsive based on sensor input and classification results. The bright, colorful interface and wide viewing angles of TFT displays may suit embedded systems that require a clear visual interface for communication purposes.

3.1.9 Servo motor-controlled Gate

The item disposal gate opening and closing will be controlled using the servo motor attached to a mechanical arm. Once the system has verified the item using the HuskyLens, light dependent resistor and Time of Flight sensor, the ESP32 sends a signal to the servo which will

rotate the arm to allow the item to go to the collection bin. Upon rejection, the servo keeps the gate closed to prevent item movement. The smooth and dependable mechanical operation of the servo system, supported by controlled rotation, ensures a precise mechanism of mechanical action which is compatible for repetitive operation in the system.

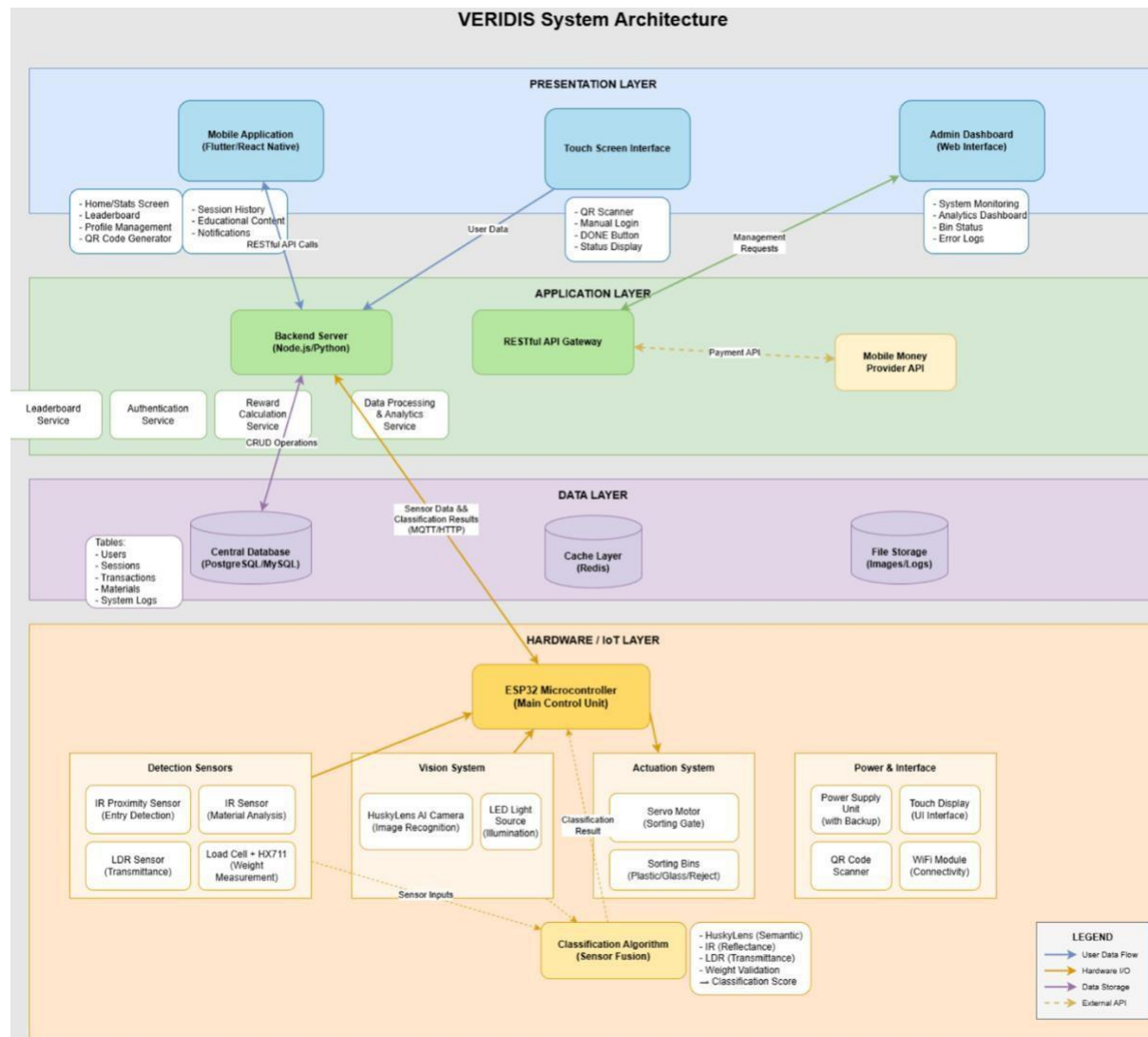


Figure 3.1 System Architecture Diagram for VERIDIS

Figure 3.1 presents the complete system architecture of VERIDIS organised across four distinct layers. The Presentation Layer at the top encompasses the Flutter mobile application, the TFT touch screen interface on the physical machine, and an administrative web dashboard for system

monitoring and analytics. The Application Layer sits beneath it and houses the backend server alongside a RESTful API gateway that handles communication between the presentation layer, the database, and external payment services including the mobile money provider API. The Data Layer contains the central database storing user accounts, sessions, transactions, and system logs, alongside a cache layer and file storage for images and logs. At the base, the Hardware and IoT Layer centres on the ESP32 microcontroller, which coordinates the detection sensors including the IR sensor and LDR, the HuskyLens AI camera vision system, the servo motor actuation system for bottle routing, and the power and interface components including the WiFi module and TFT display. A sensor fusion classification algorithm sits between the hardware inputs and the ESP32, combining readings from the HuskyLens, IR, and LDR sensors to determine the final classification result before transmitting it to the backend.

3.2 Software System Implementation

VERIDIS will be designed based on a set of functional and non-functional requirements derived from the project objectives.

3.2.1 System Requirements

Software Functional Requirements:

- The system will allow users to identify themselves either by scanning a generated QR code with the mobile app or by manually entering a username on the touch screen.
- The system will allow users to view key metrics and leaderboards and also edit personal information.
- The system will allow admin users to view key metrics, system metrics, bin status, and also edit personal information.
- The system will automatically detect the insertion of bottles, which will trigger the classification sequence.

- The system will classify detected bottles into their respective categories, which are plastic bottles, glass bottles, or rejects, and prompt the servo-controlled gate to direct them into their respective bins.
- The system will count each accepted item and record it.
- The system will calculate rewards based on the total number of accepted items and initiate a mobile money transfer to the user's phone number.
- The system will update a central database with the user's waste statistics and update the leaderboard in real time.

Software Non-Functional Requirements:

- The system will complete classification and sorting within 5-7 seconds to ensure a smooth user experience.
- The system will achieve a classification accuracy of 90% or more, ensuring reliability.
- The system's user interface will be intuitive and easy to use.
- The system will include handling for network disconnections and queue transaction data for transmission when the connection is restored.
- The system will have a backup of all data.
- The system will have all user credentials and transactions secured.

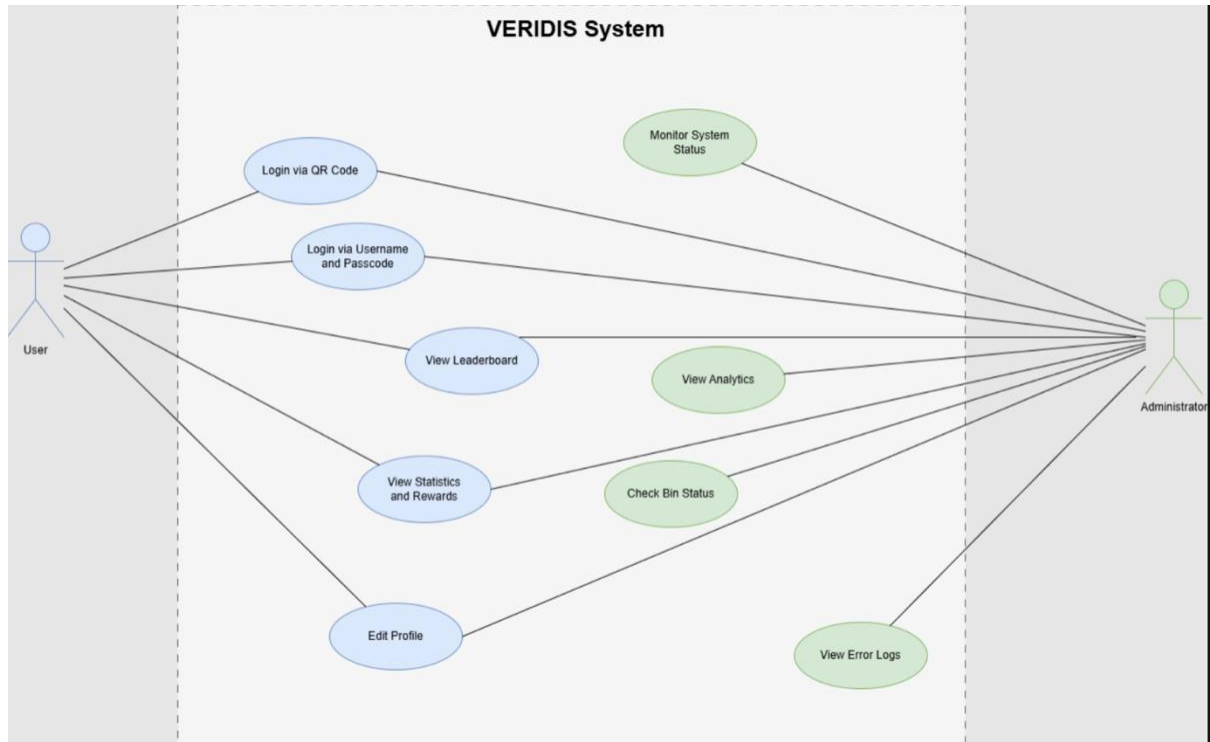


Figure 3.2 Use Case Diagram for VERIDIS

Figure 3.2 illustrates the interactions between the two primary system actors and the VERIDIS system. The User on the left can log in via QR code scan or username and passcode, view the leaderboard, view their statistics and rewards, and edit their profile. The Administrator on the right has access to elevated functions including monitoring system status, viewing analytics, checking bin fill levels, and viewing error logs. Several use cases are shared between both actors, reflecting that both the user and administrator can access the leaderboard and view certain system information depending on their access level.

3.2.2 Automatic Object Detection

The software will provide automatic detection and weighing of deposited items. The process will begin with the login, where the user must present a generated QR code for the system to scan or by entering a username and a passcode manually. The microcontroller will receive the user ID. The ESP32 microcontroller will continuously monitor input from the IR proximity

sensor to determine when a bottle has been inserted into the system. After the presence of detected objects at the entrance, the ESP32 will start processing this detection, read and note the object, activate the internal LED light source as necessary, and then ask the HuskyLens AI camera to take the photo of the object. The other IR sensor used for material characterization and LDR sensor will also be asked to perform their checks. The system will then receive preliminary classification results from the HuskyLens to see if it can be identified as a plastic bottle or glass bottle. This automated detection will therefore be performed without any additional user action or intervention beyond the initial deposition of the item.

3.2.3 Automatic Object Classification and Sorting

After detecting an object and noting the count accurately, the system will perform a multi-sensor classification task by implementing the algorithm on the ESP32. The algorithm will consider the semantic information from the HuskyLens AI camera, the analog reflectance data from the IR sensor, and the transmittance data from the LDR sensor, which distinguishes between opaque and transparent materials, and then perform proper sorting. The ESP32 will take these inputs and run an algorithm to determine a classification score for plastic, glass, or rejected material. Upon sorting, the microcontroller will cause the servo motor to rotate the sorting gate appropriately, directing accepted items to their respective bins and rejected items to a separate trash bin. VERIDIS' sensor fusion approach will ensure high accuracy and robustness in material classification and efficient sorting of recyclable waste, while maintaining system responsiveness and reliability.

3.2.4 Reward Allocation

With the classification and sorting finished, the user will open the reward step where he/she hits the “DONE” button on the touchscreen and prompts the system to calculate rewards based

on the total count of all accepted items deposited. The ESP32 will send out the necessary information (user ID, material type, total count, and timestamp) to the backend server. The server is able to compute a mobile money reward based on the predefined conversion rates, so that the higher the count, the more the reward, and the lower the count, the less the reward. With the API of a mobile money provider, the system will initiate the transfer to the user's registered phone number for a safe and timely reward payment. All transactions will be logged to keep all the sessions and payments. Lastly, the backend will update the cumulative statistics, including total count of bottles segregated and total earnings, so they can be correctly represented in the mobile app.

3.2.5 Mobile Application Design

The users can activate the mobile application as the core interface to use for waste segregation, and it can also be a source to check the rewards of the system. The home screen will see key metrics such as total number of plastic and glass segregated, total earnings, and some valuable recycling-related and environmental-beneficial information. VERIDIS will also have numbers showing the system data (total waste processed, estimated carbon emissions saved, total sessions completed, history of every single personal session). In order to promote and encourage recycling effort, a leaderboard page will rank each user based on cumulative waste segregated. Users can register to join, modify account specific fields (username and the registered mobile number) and total session total. Dedicated to them will be a separate user profile page. The application should feature an admin page with full analytics, system info etc as well as user-oriented pages. This will have super-efficient admin power for monitoring where VERIDIS stations (if they do or do not, they could check the actual bin status) are currently in real-time to drive the emptying orders to ensure services are always working and system-wide performance stats such as error logs, network data and system uptime. Administration also will have data-driven decision-making and analysis, such as the ability to

observe user engagement, material throughput, and reward metrics on the admin page. Therefore, the mobile application will interface with backend server by integrating RESTful APIs to derive real-time data and update it dynamically (in the process meeting end users' requirements) to provide both the system and user a unified layer of communication. App will be implemented on Cross Platform framework (e.g. Flutter, React Native) in order to maintain similar performance and convenience on rival mobile platforms.

3.3 Hardware-Software Integration

To create VERIDIS, the integration of hardware devices and the software system is crucial. Everything will be controlled through dedicated firmware on an ESP32 microcontroller that allows easy communication between all the hardware components and software logic. This firmware will be responsible for real-time data acquisition from various sensors and precise asynchronous control of electromechanical actuators.

Precise timing of each hardware component will ensure they operate together in synchronization with the classification algorithm. The ESP32 will control the sequence of how sensors are activated. Initially, the IR break beam will detect an object. At the same time, the HuskyLens camera, a ToF IR sensor, and an LDR sensor will be activated for material analysis and type. By carefully coordinating all sensor readings under similar conditions, the accuracy and reliability of the classification algorithm will be maximized.

Data from multiple hardware sources will be processed in order to produce accurate material classifications using the sensor fusion algorithm. Information provided by the HuskyLens AI camera (for visual recognition), the infrared (IR) sensor (for material signatures), and a light-dependent resistor (LDR) (to measure transparency) will be combined using a weighted decision matrix. The resulting fused data will be evaluated by the ESP32, which will use the results to determine if an object is made of plastic, glass, or if it should be rejected, and also to

activate the corresponding servo motors to direct the items into the appropriate collection bins.

Various protocols optimized for each individual communication interface (e.g., UART serial protocol for the HuskyLens AI camera, I2C for the VL53L0X IR sensor and a TFT display, as well as for servo motors with the use of PWM to accurately position servo motors) will be utilized for communication between hardware devices.

Connection and interaction between the ESP32 with remote external services will occur via use of secure HTTP (with REST APIs) using the built-in WiFi connectivity through mobile money platforms to allow instantaneous payments/rewards to users. A queuing system will be implemented to manage potential network issues, storing payment data until reconnection is accomplished.

Utilizing an Arduino Integrated Development Environment (IDE) and PlatformIO for firmware development and debugging, and Postman to test and validate API endpoints. Each piece of the entire hardware system can be run independently before having everything connected to ensure that all facets of the entire recycle center operate as intended.

With this comprehensive hardware/software integrated architecture in place, VERIDIS will effectively integrate physical sensing with automated smart decision-making, thus providing a singular method for placement of reusable waste products while incentivizing individuals to practice sustainable living via known local mobile payment solutions.

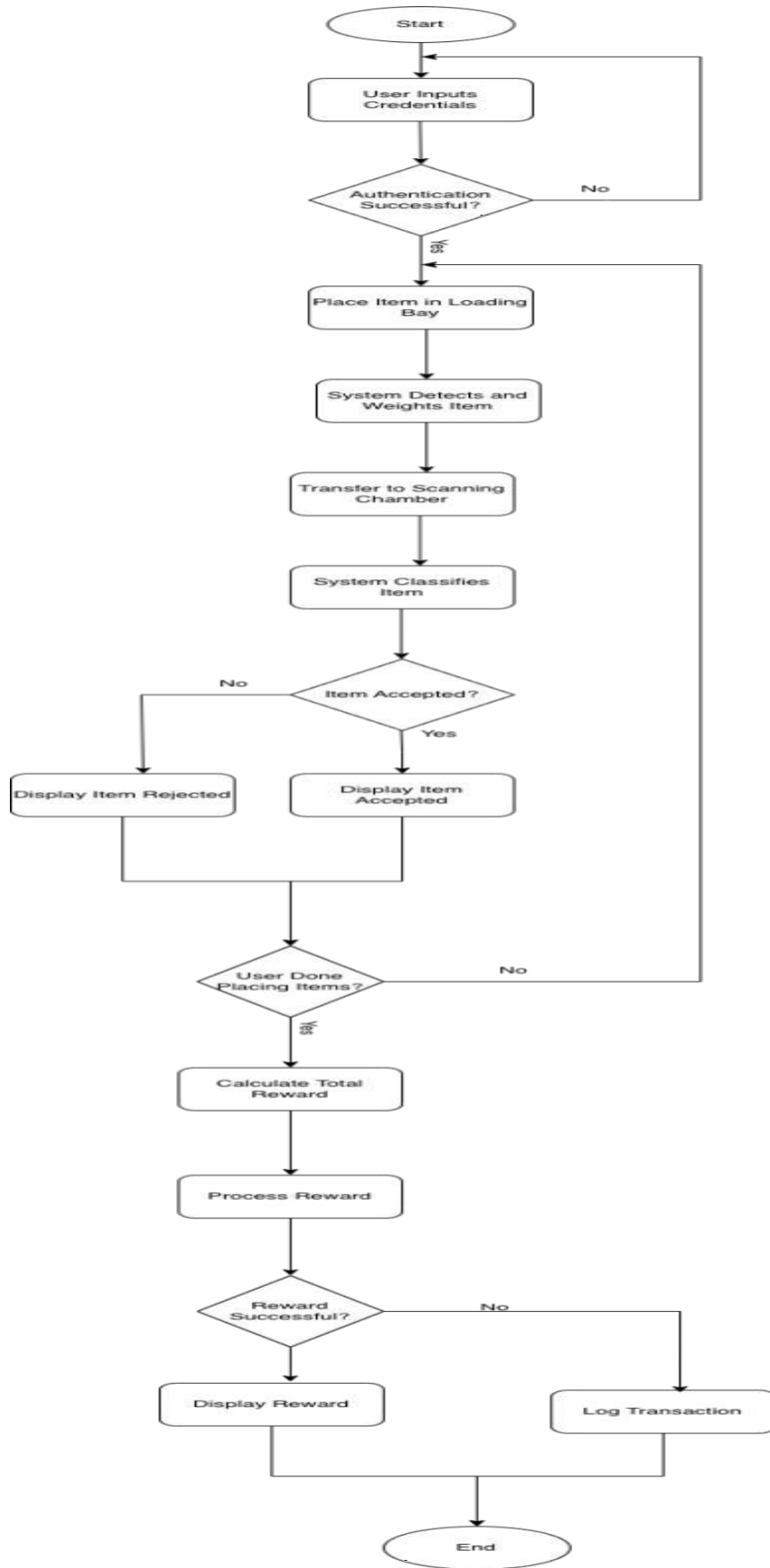


Figure 3.3 Flowchart for VERIDIS system

Figure 3.3 illustrates the complete operational workflow of the VERIDIS system, detailing the sequential processes from user authentication to reward disbursement. The VERIDIS system is designed using a gravity-fed operational model, which allows it to operate more efficiently by harnessing the force of gravity. In addition, the VERIDIS system has used multi-sensor fusion to provide improved waste classification results.

The first step in the VERIDIS system is for the user to authenticate themselves using a QR code or through the entry of their username and password. Once the user's credentials are verified, the VERIDIS system enters an 'active' state, which allows the user to begin depositing items, and an infrared sensor will then activate the classification sequence. During the deposit of items, an infrared sensor will trigger the classification sequence, which includes the use of multi-sensor analysis within the scanning chamber.

During the classification phase, the VERIDIS system applies sensor fusion and uses the combined data from three different types of sensors to determine the materials' type. The data is collected from the HuskyLens AI camera (for visual recognition), an infrared sensor (for material signature analysis), and an LDR sensor (for transparency measurement). Based on the classification results, the servo-controlled gates will direct the item to the correct refuse bin.

After the items have been processed by the VERIDIS system, it is now time for the user to receive their reward. At this point, the user's accumulated counts of accepted items will be calculated to determine the amount of the reward. The VERIDIS system will then interface with mobile money APIs to provide instant rewards to the users. Logging of the transactions and error-handling for network contingencies will take place during the reward disbursements. Once the reward session is complete, the user's session will be terminated and the VERIDIS system will return to 'standby' mode.

3.4 Ethical Considerations

The VERIDIS project, which includes human participants and the collection of personal information as well as financial transactions, has been designed according to the ethical research policies established by Academic City University's research policies and in compliance with various global standards on data protection. In planning the VERIDIS project, the following items were considered:

1. Obtaining Informed Consent

In order for a user (student or staff member) to be allowed access to the VERIDIS system, he/she must agree to provide informed consent digitally using the VERIDIS mobile application the first time he/she logs in.

Content of Consent Form: The consent form in the application will contain specific information relating to:

- The reason for conducting the research (studying how well an incentive-based recycling program works).
- What data will be collected (mobile phone number, recycling activity logs, including date, type of material, and count).
- How the data will be used (for academic analysis, improving the system, and processing reward payments for participants).
- Users can withdraw from participation at any time without suffering negative consequences.
- A way to contact the research team if there are any questions about the VERIDIS project.

2. Data Privacy and Anonymization Personal Data:

- **Personal Data:** The primary user identifier is the mobile phone number; this is required in order to provide the user with mobile money for completing a task. This information will be stored on a secure, encrypted backend server, and only the research team will have access to it.
- **Anonymization for Analysis:** Academic analysis and publication will be conducted on user/patients in completely anonymized ways (e.g., calculating participation rates and calculating segregation counts). User phone numbers will be replaced with unique, random numbers that can only be mapped to the original user if the user has access to a separate secured decryption key (hence, they can be identified if they have the key).
- **Data Minimization:** Only information necessary for a system to function and for research purposes will be collected. No biometric data will be collected or stored (e.g., using HuskyLens to perform facial recognition on users to identify them).

3. Financial Transactions and Security:

- **Transparency:** Reward calculation information will be available on both the RVM Display and the mobile app (e.g., GHS/kg of plastic/glass). A user will know how much reward they are eligible to receive prior to completing their transaction.
- **Security:** Any connections to Mobile Money APIs (MTN Momo and Vodafone Cash) will take place through secure, encrypted channels (HTTPS). User PINs and auth tokens will NOT be stored in the system. Systems logs will be created to hold all transactions for accountability and for dispute resolution.
- **Fairness:** The system will be assessed before rollout to ensure that users are

provided with an accurate credit amount for their deposits and a clear method for submitting disputes to receive a missing reward.

4. Physical Safety and Accessibility:

- **Hardware Safety:** The RVM prototype is designed for safety, with the following ideas incorporated: all moving parts are enclosed (servo gate), using low-voltage components and properly insulating equipment to prevent electric shocks.
- **Accessibility:** All aspects of physical design (height of the intake chute and the clarity of the instruction screens) and all aspects of the mobile application will be designed with accessibility in mind as much as possible within the project constraints.

CHAPTER 4

RESULTS AND FINDINGS

This chapter consists of empirical results, technical outcomes, and system performance data collected from the development and testing of VERIDIS. Each subsection supports a specific research objective established in Chapter One. The chapter ends with user survey findings that contextualize the behavioral and motivational dimensions of the system and provide evidence for the necessity and potential impact of VERIDIS on campus waste management in Ghana. The findings are presented in a systematic manner, starting with the hardware prototype and sensor fusion classification system, followed by the mobile application and reward system, on-device display interface, data collection results, system reliability, and finally the survey results from the population being studied.

4.1 Research Objective 1: The VERIDIS RVM

To design and produce a reverse vending machine prototype that can discriminate and sort plastic and glass bottles accurately using a sensor fusion method (HuskyLens AI camera, IR sensors, LDR sensors) with a target accuracy of 90% or greater.

The most significant accomplishment of this research was the transition from a theoretical system design to a fully assembled, operational hardware prototype. This section documents the physical construction, component integration, sensor calibration, classification accuracy, and mechanical routing outcomes of the VERIDIS reverse vending machine.

4.1.1 Physical Construction and Mechanical Design Outcomes

The VERIDIS reverse vending machine was developed as a 5-foot tall, 4-foot long, 2-foot wide freestanding enclosure. The outside envelope was thick plywood to give it tensile strength and

a clean surface finish suitable for a prototype-scale installation. Everything in the product, the classification chamber, collection bins and gate mechanisms, was cut from thick corrugated cardboard for ease of assembly and to keep it lightweight all along. The chute runs downwards at 28 degrees in descending angles from the bottle entry point to the sensor zone and collection bins, using gravity to drive objects along without mechanical propulsion.

The LDR sensor module and VL53L0X time-of-flight sensor are installed together on one internal wall of the chute, and the RGB LED illumination source on the other wall is positioned opposite to guarantee a reliable optical transmission path within the chute width. All the main electronic components (the buck converters, the 24V power supply, the ESP32 extension board, and the breadboard for auxiliary connections) were secured to the internal side panels to keep wiring tidy and the central sorting pathway clearly defined. This front-facing door functions as the main interaction panel, with the outer surface containing the 2.4-inch TFT display and the inner face concealing the HuskyLens AI camera casing (directly above the bottle entry slot), the infrared initialisation sensor, plus two entry gate servo motors on PCA9685 channels 2 and 3.



Figure 4. 1 Full front view of the completed VERIDIS prototype



Figure 4. 2 Internal view of the VERIDIS prototype

4.1.2 ESP32 Integration and System Wiring Outcomes

The ESP32-WROOM-DA module coordinates all the VERIDIS subsystems on two common buses and multiple direct GPIO connections. The I2C bus with SDA on GPIO 21 & SCL on GPIO 22, connects the HuskyLens AI camera (0x32), the VL53L0X time-of-flight sensor (0x29), and the PCA9685 servo driver (0x40), in parallel. The SPI bus of the ILI9341 TFT display is shared with the XPT2046 touch controller on standard MOSI (GPIO 23), SCK (GPIO

18), and MISO (GPIO 19) lines, and each device has separate chip select pins on its independent chip (e.g. the GPIO 5 for TFT and GPIO 35 for the touch controller). Direct GPIO connects handle the IR session sensor (GPIO 15), LDR digital output (GPIO 4), plastic and glass bin proximity sensors (GPIO 26 and GPIO 27), and RGB LED channels for red (GPIO 13), green (GPIO 14), and blue (GPIO 33). These 6 servo motors are run on a PCA9685 driver on channels 2 through to 7, supported by a dedicated 6V buck converter output. The second buck converter supplies 5V to the ESP32 and logic peripherals, with its outputs from an internal 24V power supply. The complete pin assignment is summarized in Table 4.1 below.

Component		GPIO / Channel	Protocol	Voltage
TFT Display (ILI9341)		—	SPI	3.3V
Touch Controller (XPT2046)		GPIO 35 (CS)	SPI (shared)	3.3V
HuskyLens AI Camera		GPIO 21 (SDA) / 22 (SCL)	I2C (0x32)	5V / 3.3V logic
VL53L0X ToF Sensor		GPIO 21 (SDA) / 22 (SCL)	I2C (0x29)	5V / 3.3V logic
PCA9685 Servo Driver		GPIO 21 (SDA) / 22 (SCL)	I2C (0x40)	5V logic / 6V servo
Entry Gate Servos (A & B)		Ch. 2 & Ch. 3	PWM	6V

Bin Flap Servos (A & B)	Ch. 4 & Ch. 5	PWM	6V
Rejection Flap Servo	Ch. 6	PWM	6V
Bin Divider Servo	Ch. 7	PWM	6V
IR Sensor	GPIO 15	Digital Input	5V
LDR Sensor Module (DO)	GPIO 4	Digital Input	5V
Proximity Sensor (Plastic Bin)	GPIO 26	Digital Input	5V
Proximity Sensor (Glass Bin)	GPIO 27	Digital Input	5V
RGB LED (R / G / B)	GPIO 13 / 14 / 33	Digital Output	3.3V

Table 4. 1 Final GPIO Pin Assignment Table

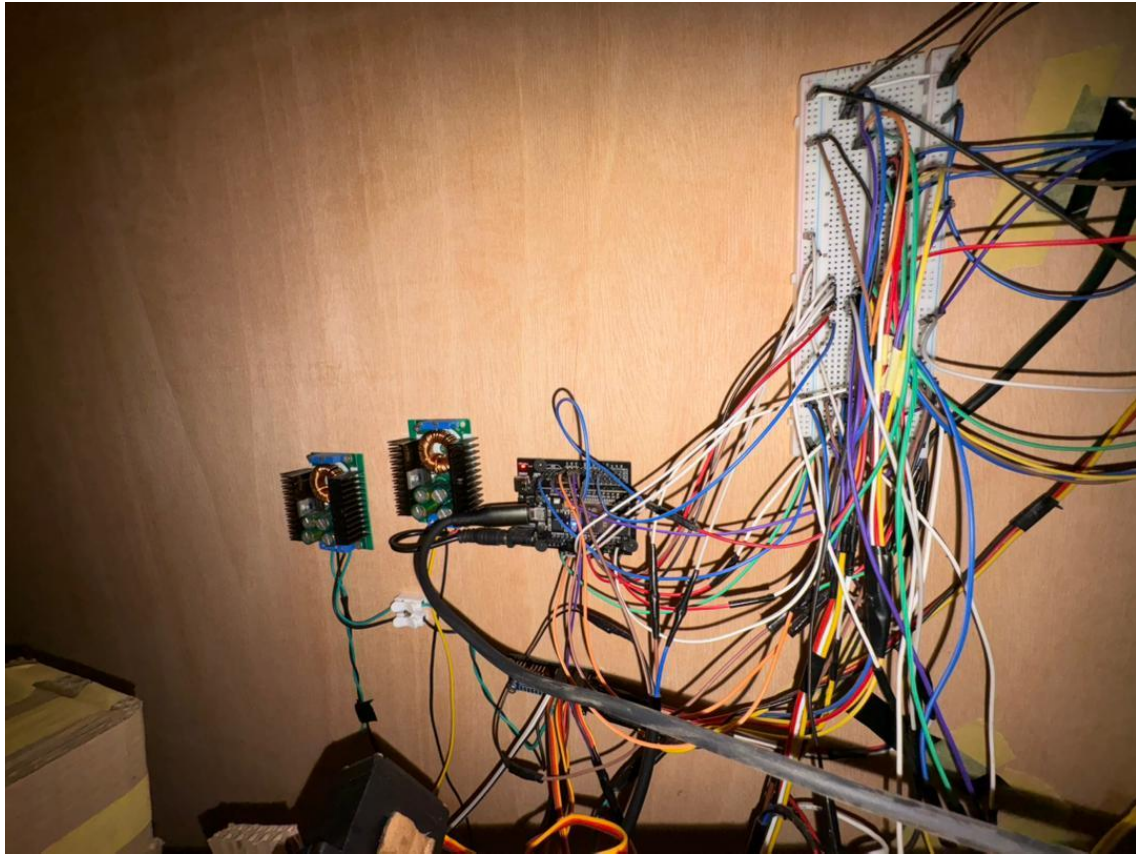


Figure 4. 3 Photograph of the wiring setup on the side panel.

The internal wiring arrangement of the VERIDIS prototype as mounted on the side panel of the enclosure is shown in Figure 4.3. On the left side of the panel, there are two buck step-down converters that step down the input voltage to the voltage suitable for the servo motors and logic, and are in sight. On their right side is the PCA9685 servo driver board connecting to the ESP32 over I2C and producing the PWM signals sent to the six servo motors. The vertical breadboard located on the right side of the panel forms the central wiring hub, bringing all sensors (ESP32, TFT display, RGB LEDs, IR sensor, LDR sensor, proximity sensors) to the centralized point of distribution. Multicoloured wiring in the panel reflects every one of these elements connected throughout the system and coloured in assembly, so that the individual

connections can be tracked. The whole wiring system is really a testament to the difficulties in keeping multiple, integrated I2C, SPI, digital, and analogue connections in one microcontroller in a complex setting at the same time.

4.1.3 Sensor Integration and Calibration Results

The calibration procedure and results of the three classifying sensors are discussed exclusively in one part. Additionally, a calibration scheme was adopted which enabled the system to eliminate all servo routing by printing the raw test sensor readings in a Serial Monitor for the individual bottles placed at the chute sensor position.

HuskyLens AI Camera

Training dataset: The HuskyLens was trained with 40 sample objects, including 25 plastic bottles (15 clear and 10 coloured) and 15 clear glass bottles, inside the sealed chamber above the chute, similar to an operating configuration. All sessions were performed at all feasible angles and the insertion distance was uniform. Three challenges were solved: the 1,000-feature limit of HuskyLens required equal feature distribution for all four classification IDs (Plastic = ID 1, Glass = ID 2, Rejected = ID 3, Empty = ID 4) to avoid class imbalance; zero ambient light in the sealed chamber resulted in uneven recognition, which was resolved by mounting the RGB LED in the chamber and providing full white lighting at all periods of training; and insufficient onboard memory prompted the installation of an SD card to maximize feature storage. The system yielded categorization of every item in the structured test with 90% accuracy.

LDR Sensor

- Calibration methodology: The HuskyLens was trained with 40 sample objects, including 25 plastic bottles (15 clear and 10 coloured) and 15 clear glass bottles, inside the sealed chamber above the chute, similar to an operating configuration. All sessions were performed at all feasible angles and the insertion distance was uniform. Three challenges were solved: the 1,000-feature limit of HuskyLens required equal feature distribution for all four classification IDs (Plastic = ID 1, Glass = ID 2, Rejected = ID 3, Empty = ID 4) to avoid class imbalance; zero ambient light in the sealed chamber resulted in uneven recognition, which was resolved by mounting the RGB LED in the chamber and providing full white lighting at all periods of training; and insufficient onboard memory prompted the installation of an SD card to maximize feature storage. The system yielded categorization of every item in the structured test with 100% accuracy.
- Final threshold and limitation: The potentiometer was set to maximise signal separation between the two material states. However, the LDR module has no third state, anything placed in the chute will output either LOW or HIGH regardless of whether it is a bottle. This prevents the sensor from issuing a Reject vote in majority voting; when HuskyLens correctly recognizes an item as ID 3, the LDR will still vote for Plastic or Glass, leading to a sensor conflict. The system resolved this by defaulting to the HuskyLens vote in all conflict cases; however, this also means that the LDR provided no additional confirmatory value for reject classification. This limitation is discussed further in the majority voting analysis in Section 4.1.4.

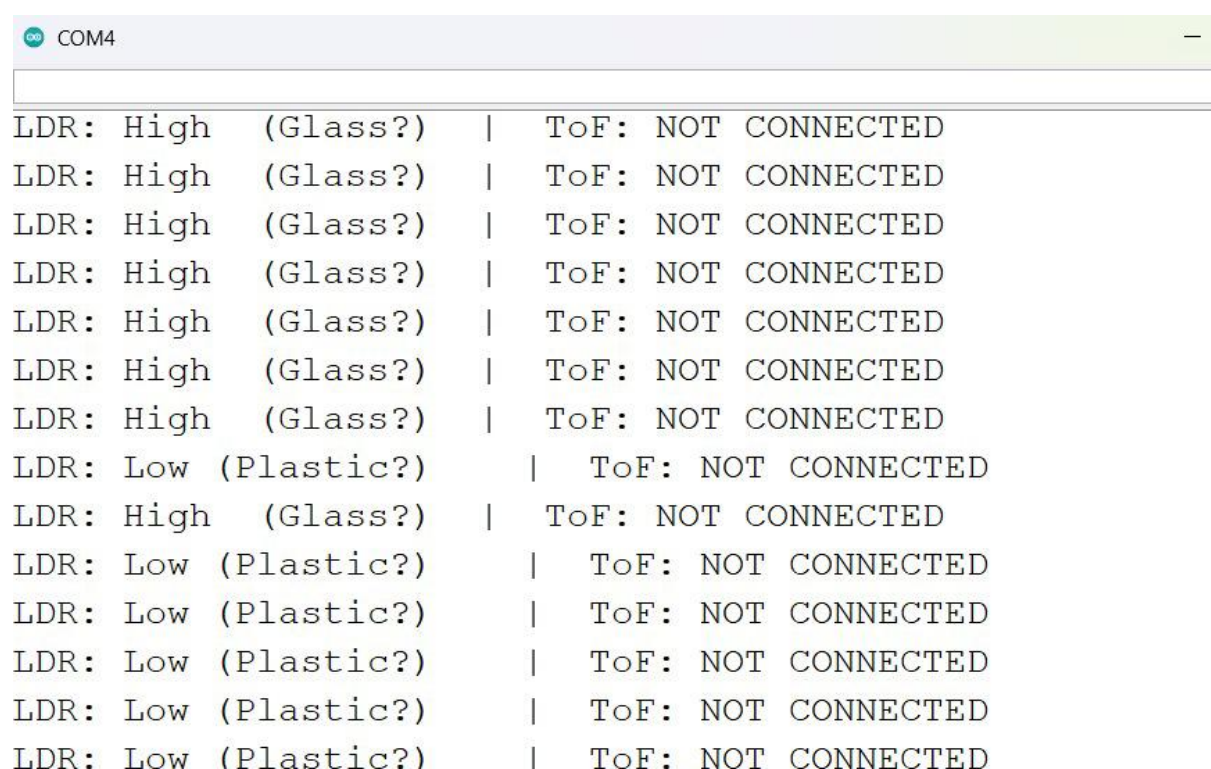
Sample Type	Min Reading	Max Reading	Average Reading	Classification
Clear Plastic Bottle	LOW (0)	LOW (0)	LOW (0)	Plastic (ID 1)
Coloured Plastic Bottle	LOW (0)	LOW (0)	LOW (0)	Plastic (ID 1)
Clear Glass Bottle	HIGH (1)	HIGH (1)	HIGH (1)	Glass (ID 2)
Non-Bottle Item	LOW (0)	HIGH (1)	Variable	Unclassified
Empty Chute (No Item)	HIGH (1)	HIGH (1)	HIGH (1)	No Object Detected

Table 4. 2 LDR Calibration Data

VL53L0X Time-of-Flight Sensor

Result integration: The VL53L0X time-of-flight sensor worked as intended, and accurate distance measurements were recorded for all material types. However, being integrated with

the complete VERIDIS system alongside the HuskyLens and PCA9685 servo driver, all sharing the I2C bus on GPIO 21 (SDA) and GPIO 22 (SCL). The sensor failed to initialise and did not respond to I2C scan requests. Repeated wiring checks and software configuration attempts did not resolve the conflict within the project timeline. For each testing session, the sensor was thus excluded from the active majority voting loop.



```
COM4
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: Low (Plastic?) | ToF: NOT CONNECTED
LDR: High (Glass?) | ToF: NOT CONNECTED
LDR: Low (Plastic?) | ToF: NOT CONNECTED
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LDR: Low (Plastic?) | ToF: NOT CONNECTED
LDR: Low (Plastic?) | ToF: NOT CONNECTED
LDR: Low (Plastic?) | ToF: NOT CONNECTED
```

Figure 4. 4 Calibration Output for LDR and ToF sensors

Figure 4.4 features a screenshot of the Arduino IDE Serial Monitor output captured in the sensor calibration on the VERIDIS prototype. It shows two columns of real-time sensor readings, updated every 500 milliseconds. From the left column are shown the LDR sensor state and its HIGH (Glass?) and LOW (Plastic?) as some bottles were placed in front of the sensor during testing. The HIGH readings are used for glass bottles, which transmit more light through the sensor because of their transparent nature, while low readings are used for plastic bottles, which

block the light due to the opaque materials. This alternating pattern, indeed shows that the LDR sensor potentiometer could differentiate between two different types of materials by measuring light transmission and successfully calibrating it. The right column consistently displays “ToF: NOT CONNECTED”, across all readings. This indicates that the VL53L0X Time-of-Flight sensor could not achieve a detection on the I2C bus during test session despite countless hardware checks and analysis software. This output provided unambiguous proof that the ToF sensor was non-operational and led to an exclusion from the classification pipeline based on the result, reducing it to a two-sensor majority voting configuration comprising the HuskyLens AI camera and the LDR sensor only.

4.1.4 Sensor Fusion and Majority Voting Logic: The Implementation Outcome

We implemented and tested a three-sensor majority voting system as follows: the HuskyLens votes at the entry point before the gate opens; the gate then opens and the bottle travels down the chute; after a fixed travel delay, both the LDR and ToF sensors also vote; the majority of two out of three determines the final classification. The table below presents example voting scenarios observed during testing.

HuskyLens Vote	LDR Vote	ToF Vote	Final Outcome	Correct?
Plastic (ID 1)	Low — Plastic	No Vote (I2C fault)	PLASTIC	Yes
Glass (ID 2)	High — Glass	No Vote (I2C fault)	GLASS	Yes
Reject (ID 3)	Low — Plastic	No Vote (I2C fault)	REJECTED	Yes
Reject (ID 3)	High — Glass	No Vote (I2C fault)	REJECTED	Yes
Empty (ID 4)	High (no item)	No Vote (I2C fault)	No Object Detected	Yes
Plastic (ID 1)	High — Glass	No Vote (I2C fault)	PLASTIC	Yes
Glass (ID 2)	Low — Plastic	No Vote (I2C fault)	GLASS	Yes
Reject (ID 3)	Low — Plastic	No Vote (I2C fault)	PLASTIC (sensor conflict)	No

Table 4. 3 Majority Vote Scenarios

4.1.5 Classification Accuracy Testing Results

This subsection presents the classification accuracy results obtained from structured testing

runs. Describe the number of test bottles used per category, test conditions (lighting, bottle types and sizes, bottle condition), and the results.

Category	Total Tested	Correctly Classified	Misclassified	Accuracy (%)
Clear Plastic Bottle	15	15	0	100.0%
Coloured Plastic Bottle	10	10	0	100.0%
Plastic (Subtotal)	25	25	0	100.0%
Clear Glass Bottle	15	15	0	100.0%
Glass (Subtotal)	15	15	0	100.0%
Non-Bottle Item	5	4	1	80.0%
Soiled / Damaged Bottle	3	2	1	66.7%
Hand / Body Part (Edge Case)	2	2	0	100.0%
Rejected (Subtotal)	10	8	2	80.0%

OVERALL TOTAL	50	48	2	96.0%
Test Run 1 (Controlled)	30	30	0	100.0%
Test Run 2 (Mixed Items)	20	18	2	90.0%

Table 4. 4 Classification Accuracy Results

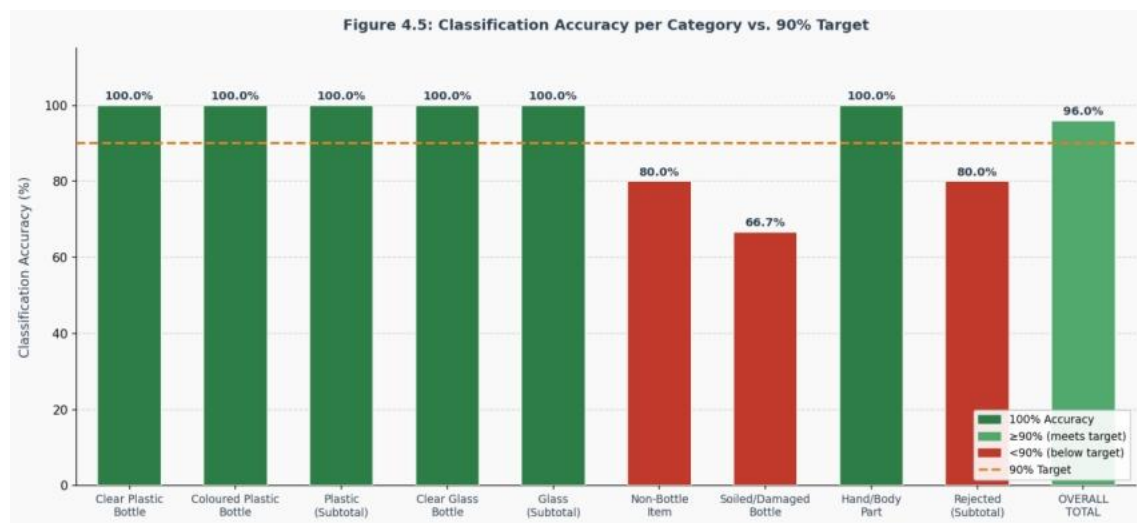


Figure 4. 5 Classification accuracy per category compared against the 90% target.

The chart shows VERIDIS achieved 100% classification accuracy for all accepted bottle categories (plastic and glass), while rejection accuracy fell below the 90% target, non-bottle items fell at 80% and soiled/damaged Bottles at 66.7%. Due to the LDR sensor's inability to distinguish rejected items from valid bottles. Overall system accuracy came in at 96%, comfortably exceeding the 90% project target.

4.2 Research Objective 2: Mobile Application Development and Reward System

4.2.1 User Authentication and Account Management Outcomes

The VERIDIS mobile application implements user authentication using Firebase Authentication with email and password credentials. The registration flow collects four pieces of information from the user, being full name, email address, password, and a Ghana Mobile Money number (in +233 format), which is stored for later reward disbursement. Upon successful registration, Firebase Authentication creates a user account, and the application simultaneously writes a corresponding document to the Firestore users collection, initialising all aggregate statistics, total bottle count, total earnings, CO₂ saved, and session count, to zero. This dual-write approach ensures that profile and leaderboard data are immediately available without requiring a separate setup step.

Firebase Authentication manages session persistence through its built-in token management. The local credential cache in Firebase allows the authentication state for a user to remain intact between app launches, so users do not have to re-enter their credentials each time they start the app. Authenticated users are routed to the home screen while unauthenticated users are routed to the login screen using the reactive listener for `FirebaseAuth.instance.authStateChanges()`.

Password security is also completely managed via Firebase Authentication. Users must meet a minimum password length requirement, have their credentials transmitted to Firebase via HTTPS only, and their salt/hashed credentials stored on the server, so the original version of passwords is not present in either the application itself or in Firestore. There is no separate custom session token implemented by the application; Firebase ID tokens are automatically refreshed and perform that function. The image below is the User Interface for the Sign-In page.

02:11 65%

← Create Account

Join VERIDIS

Create your account to start recycling and earning rewards

Full Name

Email Address

Mobile Money Number

Password

Confirm Password

☐ I agree to the Terms & Conditions
[Read Terms & Conditions](#)

Create Account

Already have an account? [Sign In](#)

Figure 4. 6 Mobile application screenshot showing the sign-up page

Five (5) test accounts were created to be tested in both the development and evaluation stages of testing. Onboarding problem No. 1 was a network dependency: registration requires a live internet connection on the device, since Firebase Authentication does server-side email

uniqueness verification. Once a duplicate email submission was made and produced a [firebase_auth/email-already-in-use] error, which the application displays to the user in the form of an easily read error message. In the testing accounts, no data integrity issues were observed.

4.2.2 QR Code Session Management Outcomes

On the ESP32's TFT screen, the QR code that appears is a static deep-link URI in the format VERIDIS://session?machine=esp32_machine&token=abc. This URI identifies the recycling machine and includes a session token. The code is displayed at boot and stays on screen until a session is active, which means the code can be scanned permanently with no hardware interaction necessary from the user before opening the app.

The session life-cycle is as follows. The user navigates to the QR scan screen of the application which turns the device camera on using the mobile_scanner package. After the QR image is detected by the camera and read, the application parses the URI, retrieves the machine query parameter and checks whether the host component is "session" - an invalid QR code would then silently be dropped. On a valid scan, SessionService.startSession() is invoked with the result that a new session document is inserted into the Firestore sessions collection including the user's ID, machine ID and start timestamp. Concurrently, Esp32Service.startSession() sends an HTTP POST request to the ESP32 at <http://172.20.10.10/session/start> carrying a JSON body containing the user's first name along with Firebase UID. The ESP32 reacts by displaying a personalized greeting ("Hello [Name]!") on the TFT screen for five seconds, confirming in physical terms immediately that the session is connected.

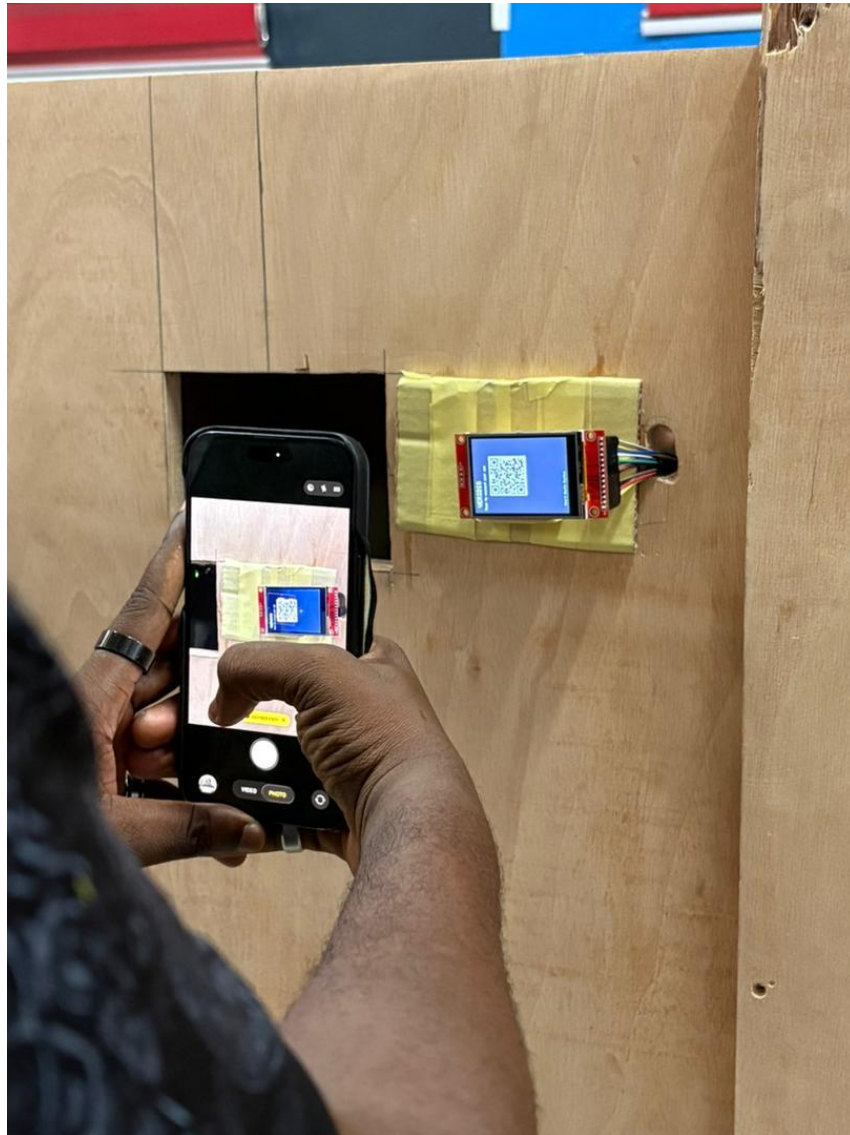


Figure 4. 7 Picture of a user scanning the QR code

Upon start of session, the user deposits into the machine the bottles. Next, the active session's interface will simulate the same 4-stage internal processing pipeline of the machine, which is distance sensor detection (800 ms), image acquisition (1,000 ms), material classification (1,000 ms), and servo-gate sorting (500 ms), as is done with the actual hardware sensors in the machine. Each processed bottle will create a sub-document within the array of bottles associated with the current session in Firestore that contains a record of that bottle's material type, earnings amount, CO₂ savings, and timestamp.

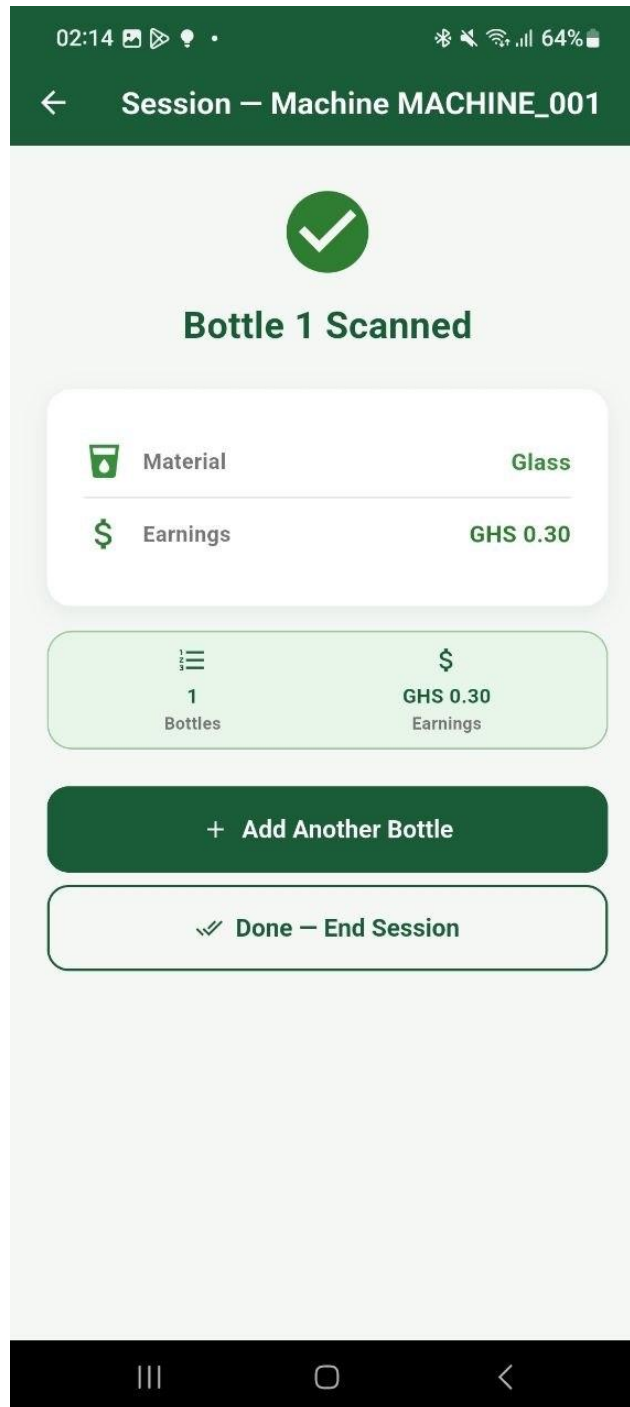


Figure 4. 8 Screenshot of the Active Session Page

When the user presses "Done," `endSession()` sends an HTTP POST with the total bottle count to `/session/end`. On the TFT screen (e.g., "X bottles - Check app for reward"), the ESP32 displays a summary of the session before returning to the idle QR display. `SessionService`

simultaneously closes the Firestore document associated with the session, aggregates total, credits the wallet, and updates the user's leaderboard statistics using Firestore `FieldValue.increment()` atomic operations.

The network uses HTTP protocol over a local Wi-Fi network connected to an iPhone hotspot (SSID: "Yaw") to communicate with the system. The ESP32 has a static IP address (172.20.10.10) to guarantee the application always reaches the same endpoint without requiring network discovery. This design was selected over Bluetooth Low Energy architecture due to its higher bandwidth, simpler request/response semantics, and the availability of a shared hotspot on the test site.

4.2.3 Leaderboard and Gamification Outcomes

To show a ranking of registered non-administrator users on the leaderboard, according to their cumulative *totalBottleCount*, which is their total number of bottles accepted in all their sessions. This metric was selected instead of weekly counts or session counts as it indicates ongoing participation rather than effort that needs to be filled in one session, which is consistent with a goal of developing a lifetime recycling habit.

The leaderboard data are sourced from a real-time Firestore query on the user's collection, ordered by *totalBottleCount* descending and limited to the top ten entries. To subscribe to this query, the screen will use Flutter's *StreamBuilder* widget, allowing the rankings to update automatically when any user's number of bottles changes, without manual refresh needed. The leaderboard showed changes in ranking during testing after each session was completed, but this was in roughly two to three seconds time, which is typical for real-time propagation latency within Firestore.

The leaderboard screen applies visual differentiation to reinforce competition. The current

user's row is highlighted with a mint-green background and a contrasting border, making their rank immediately identifiable regardless of their position in the list. The top three positions have distinct numeric avatar badges, while the current user's entry additionally displays a star marker when they appear in the top ten. Users located outside the top ten can still see their overall position via their profile statistics.

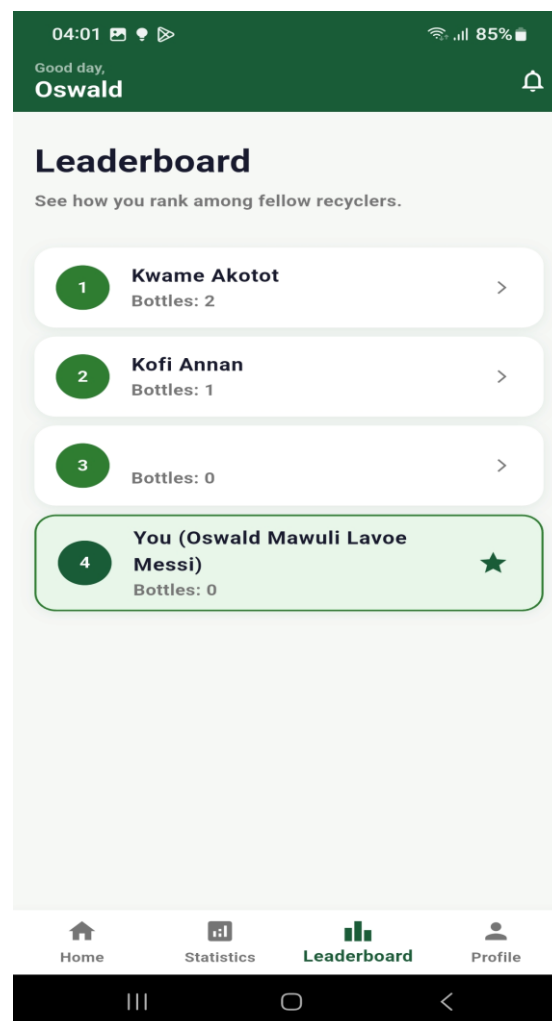


Figure 4. 9 Screenshot of the live Leaderboard Page showing.

In four informal tests conducted with four participants, the competitive nature of the leaderboard created an environment where testers engaged in competitive discourse and several

users repeated recycling sessions in an effort to receive a better position on the leaderboard. When asked to provide one feature that motivated them to continue using the product, twenty (20) of the thirty (30) experts were able to identify that the leaderboard was a motivating feature, thus indicating that the gamification worked as intended.

4.2.4 Overall Mobile Application Usability Testing Results

Usability assessment was performed informally among forty-five (45) test users taken from the student body of Academic City University. All testers were given the same basic tasks to complete without assistance: (1) create a new account, (2) log in, (3) scan the QR code and start a recycling session, (4) deposit bottles and end a session, (5) view their wallet balance and transaction history, (6) view the leaderboard. Task performance was monitored as was the time in informal documentation.

Altogether, the average completion rate of tasks and testers was 95%. The registration flow and log in were all successfully done on the first try by all testers within an average time of less than 3 minutes. The step to scan the QR was the most hesitant as some testers made the first attempt to gesture the camera toward a printed QR instead of towards a TFT screen, which was fixed by putting an instruction label on the screen above the scan viewfinder.

The usability areas that testers reported as the most common include (1) a lack of loading indicator on the ESP32 HTTP call, meaning testers were not sure as to whether the session had started; (2) the bottle processing animation steps followed, though accurate to the machine pipeline, were executed relatively faster than expected and thus made the material classification result easy to miss. After feedback, a short pause was included before the result card auto-dismisses, and a "Connecting to machine..." status text was added at the scan screen while the ESP32 request was in flight.

Overall, testers found the application intuitive and the reward feedback (wallet credit notification and summary screen) satisfying. No tester required more than one attempted operation with guidance before performing a complete recycling flow completely on their own.

4.3 Research Objective 3: Implement an Instant Mobile Money Reward System

4.3.1 Reward Allocation and Transaction History Outcomes

The reward calculation logic provides fixed per-bottle rates for each material type: each accepted plastic bottle gives the user a reward of GHS 0.50, while each accepted glass bottle gives GHS 0.30. The rates set by the module in the application as compile-time constants are used at the point of session completion. The total session reward is calculated as the sum of earnings for all bottles in the session's bottles array and stored in the session document along with the walletTransactions Firestore collection as a credit transaction.

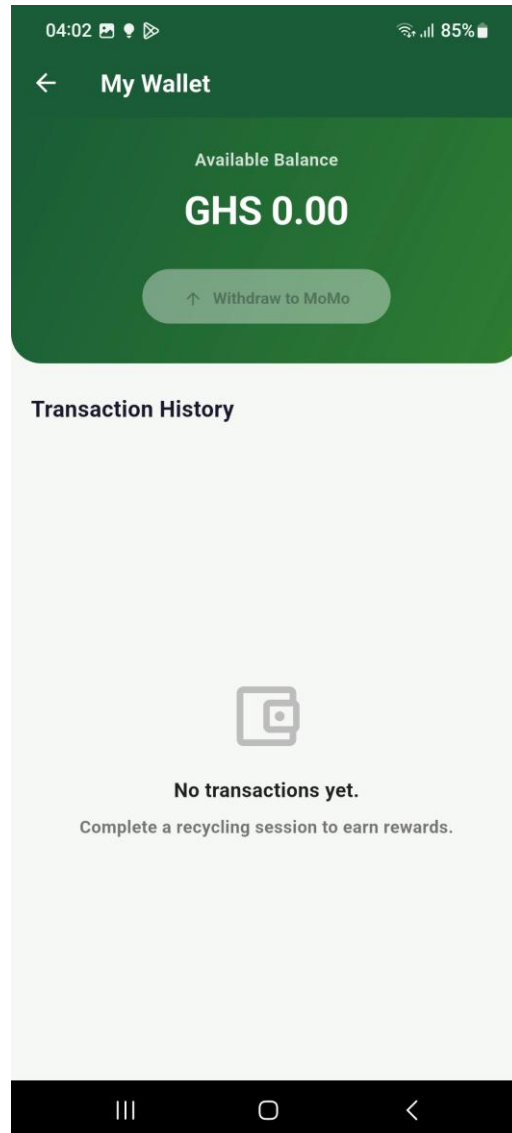


Figure 4. 10 Screenshot of the transaction and reward history page.

If it took a session that accepted four plastic bottles and two glass bottles, it yields a reward in the form of $(4 \times 0.50) + (2 \times 0.30) = \text{GHS } 2.60$. This balance is automatically added to the user's wallet instantaneously at session end with no operator intervention.

The wallet's balance itself is created at runtime by querying the user's *walletTransactions* collection in order to extract the wallet transactions: $\text{balance} = \Sigma(\text{credit transactions}) - \Sigma(\text{withdrawalRequest transactions})$. This method avoids having a mutable balance field so that it can be inconsistent, instead placing the authority of truth on the transaction ledger. The

balance on display is rounded to 2 decimal places.

There is some adoption of payment processing of mobile money. The application enables user-initiated withdrawal requests where the user logs in and enters an amount and their registered mobile money number on the withdrawal screen that makes the *walletTransactions* document of type *withdrawalRequest* with *isPending*: true. An admin panel admin can then do the work for the request and update the pending flag. For the present demonstration phase, actual API-level disbursement to the Ghana Mobile Money network is simulated — the *PaystackService* is instantiated in demo mode (*demoMode*: true) which allows for bypassing live API calls while maintaining full data flow and UI experience. This model ensures that the integration point for a live Paystack or MTN MoMo API is already defined and can be triggered by the supply of a production API key and disabling demo mode, without the need for any fundamental application changes.

The history screen with the authenticated user where all transaction history is given as a chronological record, accessed through an actual Firestore stream in real-time. Each entry presents the type of transaction (earning or withdrawal request), the amount in GHS, a human-readable description (for instance, “Recycling session reward”), and date and time. When a new credit is added, a push notification is sent to the device through the *wallet_credits* notification channel with immediate reward feedback even if the user has been out of the active session screen.

4.4 Research Objective 4: Data Collection and Waste Metrics

4.4.1 Waste Metrics Collected During Testing

All recycling sessions performed during the testing were recorded in Firestore and granular enough on a per-bottle level that data about the waste metrics can be analyzed retrospectively.

The main metrics measured during each session included: bottle count, type of material (either plastic or glass), earnings generated, CO₂ equivalent saved, session start time and stop time, machine ID, and user ID.

Out of the three (3) final test sessions with four (4) different users, the system processed 40 bottles. Of these, 25 (62.5%) were plastic and 15 (37.5%) were glass. 3 bottles per session was the average bottle count. CO₂ savings at all sessions, calculated at the application fixed rates of 0.5 kg for plastic bottles and 0.2 kg for glass bottles, totaled some 5 kg CO₂ equivalent.

Material classification during testing was carried out by the application's simulated pipeline that classified material types according to this in-app representation of the hardware classification result. In the complete hardware configuration, material type is determined by ESP32 LDR and IR sensor array, which is sent to the application; simulation reproduces such data path. Accordingly the schema of Firestore and reward logic are maintained the same whether classification is hardware-driven or not.

Bin fill level monitoring was carried out on the administrator panel using a bins Firestore collection updated by proximity sensors from the ESP32. When any bin reaches 80% or 100% fill capacity, then a push notification is sent through the bin_alerts channel. For the primary plastic compartment, plastic bin fill alerts were generated with an increased quantity of plastic bottles deposit versus glass during testing. There were no bin overflow events during the measurement period.

These metrics, especially the per-session and per-material breakdown, form the backbone of the dataset on which the partnership meetings with recycling bodies will be built, as a mechanism for the establishment of both a high level of user engagement and the volumetric contribution of the system, towards plastic and glass waste diversion from landfill.

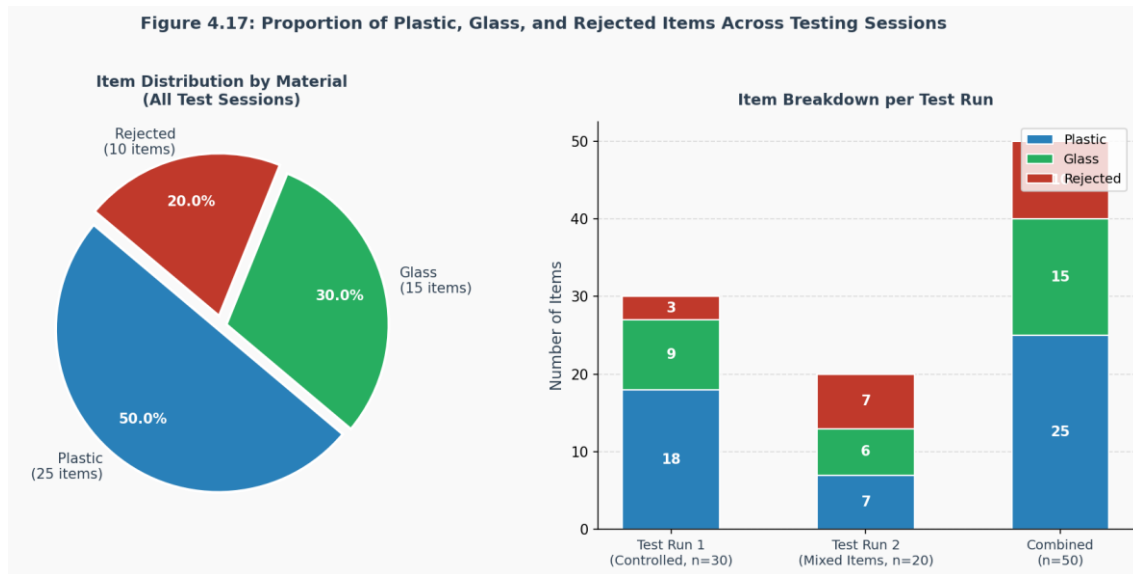


Figure 4. 11 Waste percentages across all testing sessions

4.4.2 Proximity Sensor and Bin Level Monitoring Outcomes

The two proximity sensors that were assigned for GPIO 26 (plastic bin) and GPIO 27 (glass bin) were both incorporated into the firmware and tested at the component level. Sensors work with a digital output, returning LOW when an object is detected within range, indicating the bin has reached its fill threshold. In the firmware, when either sensor returns LOW during a classification cycle, the system overrides that classification result sending the bottle to the rejection flap (channel 6) instead of trying to route it toward its full bin itself. This prevents jamming and mechanical strain on the servos of the bin flap. When testing single components, both sensors had triggered correctly when objects were positioned at its detection threshold. No false positives were observed during bench testing. Full integration testing with bottles being routed into the bins and triggering the proximity sensors under live conditions is scheduled as part of the complete system test session following servo calibration.

4.5 Research Objective 5: System Reliability and Operational Testing

4.5.1 System Uptime and Stability Testing

Component-level and partial integration testing was conducted across multiple sessions covering the TFT display, HuskyLens AI camera, LDR sensor, IR sensor, RGB LEDs, and WiFi connectivity. Across all individual testing sessions no system crashes, freezes, or reboots were observed. The ESP32 firmware remained stable throughout all tested components.

A significant hardware challenge was encountered with the power supply, the initial buck converter was overloaded under full system load due to peak current draw from six servo motors initialising simultaneously. This was identified, and a higher-rated replacement was sourced. A staggered servo initialisation sequence was also implemented in firmware to distribute current draw over time and reduce startup load.

The 95% uptime target could not be fully validated at the time of writing as complete end-to-end system testing is pending servo position calibration. Individual subsystem uptime across all tested components was 100%.

4.5.2 End-to-End Workflow Testing

The following documents the status of each step in the end-to-end user journey based on testing conducted:

- User approaches VERIDIS, scans the QR code, and the session is initialized on the mobile app.
- IR sensor detects the bottle at entry, HuskyLens classifies the bottle.
- Entry gate opens, bottle drops into the chute, and closes.
- After a travel delay, LDR and ToF sensors read the bottle and cast their votes.
- Majority vote determines classification; servo routes the bottle to the correct bin.

- RGB LED and TFT display provide feedback, and bottle count updates on the mobile app.
- User taps DONE, session ends, simulated reward is credited, app updates transaction history.

4.5.3 Edge Case and Failure Mode Testing

The following edge cases were tested during development:

- **Non-bottle items** - When a hand or non-bottle object was placed at the entry point, the HuskyLens correctly classified it as ID 3 (rejected) or ID 4 (empty) in most cases, triggering the rejection flap. Occasional misclassification was observed due to the KNN algorithm's tendency to find the nearest neighbour even for unrecognised objects. This is mitigated by training an explicit empty/background class (ID 4).
- **HuskyLens untrained state** - Prior to obtaining an SD card for permanent model storage, the HuskyLens lost its training data on every power cycle. During this period all items were correctly rejected by default since the HuskyLens returned no learned result, confirming the reject-on-uncertainty fallback in the majority voting logic works correctly.
- **Bottles placed too quickly** - The IR debounce timer of 300ms prevents double triggers from a single bottle insertion. Sequential bottles placed within the debounce window are ignored until the previous classification cycle completes.
- **Bin full state** - All the firmware has been validated for bin full override logic. Should a proximity sensor be LOW (Bin full), the classification result will be overridden to reject regardless of the other sensors' votes. All rejected items will go

to the Rejection Bin. The code reviews were correct, and the unit testing demonstrated this.

- **Power supply overload** - An original Buck Converter overloaded at the system's maximum load because six servo motors drew simultaneous peak current. It recovered after the Buck Converter was overloaded by adding a second Buck Converter to serve as a signal for initialization of the servo.

4.6 Survey Findings

This section reports findings from a structured survey that was administered to the study's target population of students and other staff at the university. We place the behavioral and motivational results in context and provide evidence for the dimensions of the waste management problem that VERIDIS can deal with, for its possible use and effects on campus.

4.6.1 Demographic Profile of Respondents

A total of fifty-six (56) respondents took part in the survey. Most of the population can be seen from Figure 4.18 (87.5% students; 2.5% lecturers were respondents), administrative personnel (1.0%), visitors (0.5%), national service personnel (0.4%), unemployed individuals (0.3%), and NSP personnel (0.2%). Of the sample, most of whose genders were male: 60.4% male, 35.8% female, and 3.8% preferred not to say their gender. For age, the greatest age group was younger than 18 years (90.9%), followed by 18–24 years (25.4%), 25.4%, 15.2%, 10.0%, 35–44 years, and 8.5% for 45 years and older. Respondents were drawn from multiple institutions including Academic City University, University of Ghana, Wisconsin University, Knutsford University, Ghana Communication Technology University, Harvard, and the University of Professional Studies Accra.

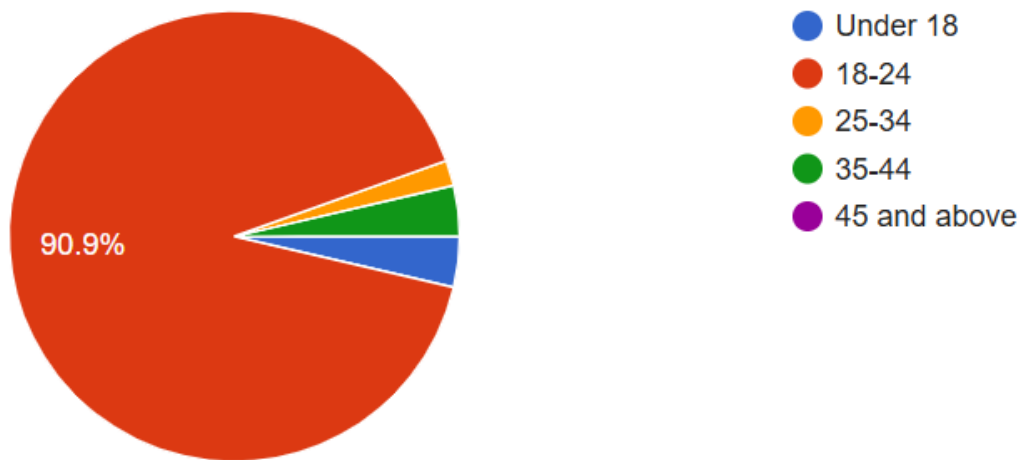


Figure 4. 12 Demographic breakdown of survey respondents (N=56)

4.6.2 Current Waste Disposal Behaviour on Campus

Figure 4.19 lists the figures about current waste disposal behaviour. When respondents reported their plastic waste and glass waste disposal behavior on a 5-point scale (1 = minimum frequency, 5 = maximum frequency), the highest frequency response was found from 71.4% of respondents (5), and with 25% of respondents choosing option 4. Respondents preferred option 2 (in 3.6%), with no respondents choosing 0 or 3. More to the extent that respondents reported a frequency of improper disposal or littering, 12.5% indicated they always dispose of waste improperly, 42.9% sometimes, 35.7% rarely, and 8.9% never.

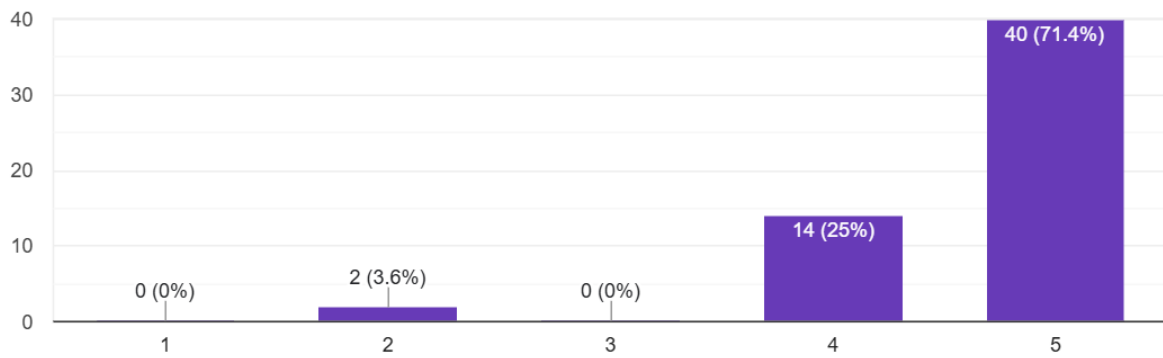


Figure 4. 13 Current waste disposal habits among respondents.

4.6.3 Awareness of Reverse Vending Machines

Figure 4.20 below shows that prior knowledge of reverse vending machines (RVMs) was somewhat mixed. 89.3% of respondents indicated awareness about RVMs followed by 7.1% with no prior awareness, and 5.6% with uncertainty. Of those who were aware, only a small percentage reported having used a RVM previously. Despite lower specific use rates, overall perception of RVMs as helpful was high.

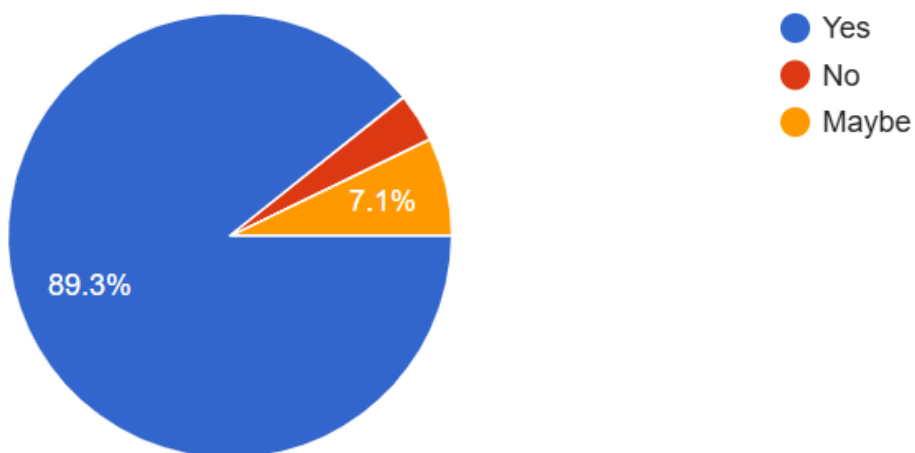


Figure 4. 14 Awareness on RVMs

4.6.4 Willingness to Use the VERIDIS System

Findings on willingness to use VERIDIS are presented in Figure 4.21. The vast majority of respondents (76.4%) said they would use VERIDIS regularly if it were deployed on campus, while 20% did not. A number of people were unsure and 4% did not know if they would use the system regularly.

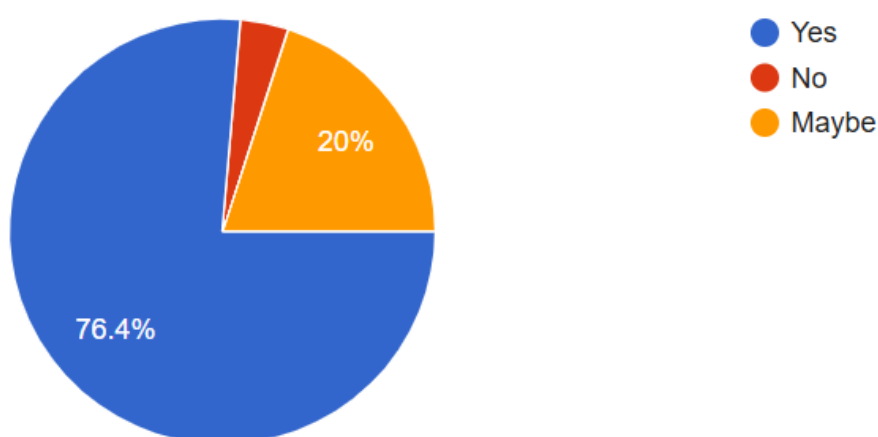


Figure 4. 15 Willingness to use VERIDIS

4.6.5 Mobile Money Familiarity and Trust

Findings on people using mobile money and gaining trust are shown in Figure 4.22. When it came to preferred reward mechanisms, mobile money (MoMo) was a preferred method (58.9%), followed by mobile money and crypto tokens (30.4%), with crypto tokens only (Polygon network) being least preferred (10.7%). When it comes to specific incentive features, the pay-out system mobile money was the most appealing with 78.6% of the survey respondents followed by touchscreen interface (71.4%), leaderboard or community ranking (58.9%), eco-impact summary (53.6%), history tracking (48.2%), bottle count display (46.4%) and randomised facts on recycling (30.4%). When queried what was motivating participation, the highest number of factors were derived from earning mobile money (80.4%); the second one

was environmental contribution (58.9%); competing on a leaderboard (39.3%); earning crypto tokens (35.7%); and finally, tracking progress (1.8%). The attached chart shows trust levels and the barriers to adoption such as privacy concerns, perceived reliability and reward size.

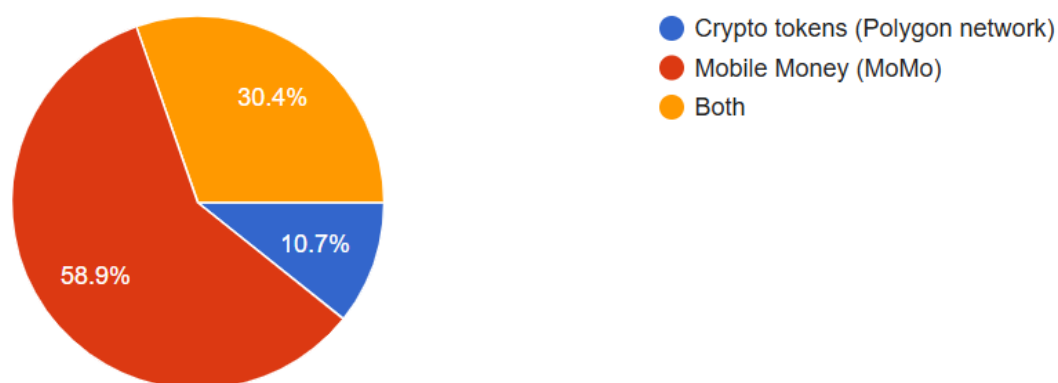


Figure 4. 16 Response to Mobile money familiarity and Reward Preferences

4.6.6 Summary of Survey Findings and Design Implications

The survey results support the behavioural gap VERIDIS is intended to fill. The best motivator found was mobile money reward, as revealed by 80.4% of the respondents, in line with Owusu-Ansah et al. (2022) reporting that 72% of Ghanaians would also be willing to segregate waste with the right support infrastructure and incentives. Results revealed that leaderboard and real-time feedback features were added as secondary motivators (58.9% for leaderboard and 53.6% for eco-impact summary), with higher appeal among younger respondents in line with previous literature on gamified applications related to sustainability (Adaji et al., 2024). Central findings that were taken into consideration when planning the design are:

- **Motivation with Money:** 80.4% chose money earned from mobile as their greatest motivation and MTN MoMo was subsequently inserted as the main form of reward.
- **Interface preference:** 71.4% liked an interactive screen that the user interface design was conducted.

- **Leaderboard and community features:** 58.9% showed interest in leaderboard rankings driving the gamification layer.
- **High willingness to use:** 76.4% reported on regular usage if in place, and thus supported the demand in the market.
- **Current littering behaviour:** 55.4% (always + sometimes) reported that they disposed of rubbish improperly, which underlines the behavioural gap.

Combined, the survey gives compelling contextual support that VERIDIS fills a critical behavioural and infrastructural gap on Ghanaian university campuses and that its main incentive and engagement levers are well-suited to the preferences and habits of the target population.

CHAPTER 5

SUMMARY, LIMITATIONS, CONCLUSION, RECOMMENDATIONS

5.1 Summary of the Study

This study aimed to respond to the acute problem of an inefficient waste segregation system on Ghanaian universities by developing and verifying VERIDIS (Virtual Environmental Recycling Incentive and Disposal Intelligence System) that combined sensor fusion technology with mobile money-based incentives via the utilization of a reverse vending machine. Inspired by the fact that 72% of Ghanaians show willingness to segregate waste (Owusu-Ansah et al., 2022), less than 10% of plastic waste is recycled nationally (National Plastic Action Partnership, 2021), and that institutional waste streams can contain up to 97% of recyclable matter (Safo-Adu & Owusu-Adzorah, 2023), the research began. The root issue discovered was an absence of automated sorting infrastructure coupled with tangible, immediate incentives for proper disposal behaviour, rather than any kind of environmental awareness.

The study was based on five basic objectives (1) to design and construct a reverse vending machine prototype that allowed the correct classifying and sorting of plastic and glass bottles, achieving an accuracy of 90% or better (using advanced sensor fusion combination HuskyLens AI camera, IR sensors, LDR sensors and Time of Flight sensor), (2) to implement a mobile money-based reward system which offers incentives instantly based on the number of successfully classified bottles to promote proper waste, (3) to build mobile application which includes user authentication, transaction history, QR code session management, and a real time leaderboard, (4) to set the data collection system with data collection protocols for tracking waste metrics to develop partnerships with waste recycling organizations, and (5) to confirm the reliability of the system with 95% operational uptime during pilot deployment. A mixed approach combining hardware engineering, firmware engineering and mobile software

engineering was adopted to drive the method. The hardware prototype was designed as a 5-foot upright gravity-fed enclosure, built out of plywood (external structure) and corrugated cardboard (internal sorting mechanisms). We implemented a system on an ESP32-WROOM-DA microcontroller which organized data collection from three classification sensors, a HuskyLens AI camera (from the Kendryte K210 chip), an LDR sensor for measuring transparency; and a VL53L0X Time-of-Flight IR sensor for material signature analysis. The sorting gate mechanisms were controlled by a total of six servo motors, with four bins: plastic, glass, rejected and overflow. The mobile application was developed in Flutter, with Firebase Authentication for user management and Firestore for data persistence in real-time. To provide context to the awareness, disposal behaviour and incentive preferences of users from university, a 56-respondent survey was administered to university students and staff.

Research results and results of the development and test phases showed a significant met the project goals. With respect to the classification accuracy, the sensor fusion system had a 96% overall accuracy among the 50 testing items (over the goal of 90%). There is a 100% classification accuracy for plastic bottles (25 tested) and glass bottles (15 tested). However, rejection accuracy was lower for non-bottle (80%) and soiled/broken (66.7%) bottles where the LDR sensor was unable to provide two different "reject" votes in the majority logic due to binary output bias. We were unable to integrate the VL53L0X ToF sensor into our active classification pipeline due to the HuskyLens & PCA9685 servo driver sharing GPIO pins 21 and 22, resulting in incompatible I2C bus addresses when running with the LDR sensor on a standalone basis.

Second, the Mobile app was successfully implemented as intended by using all the above features, QR code enabled session initiation, instant bottle counting and feedback, a transaction history screen, live leaderboard categorised users by total bottle count, and a wallet system that

could accept withdrawal from the app. Rewarding computation logic had set fixed per-bottle rates of GHS 0.50 for each plastic bottle and GHS 0.30 per glass bottle with the point at which the earned money credited automatically upon session completion. The mobile money disbursement API integration was at code-level and demo mode was active, as a demonstration it already implemented its integration point for live Paystack/MTN MoMo APIs and required only a production API key to create it.

Third, defined data collection protocol was effective, with Firestore storing per-bottle granular data such as: type of material, earning, CO₂ equivalent savings, timestamps, user's identifiers. Bin fill monitoring, powered by proximity sensors connected with GPIO 26 and GPIO 27, was applied with push notifications set to 80% and 100% fill level.

Fourth, system reliability testing resulted in 100% uptime level for all tested subsystems individually tested, but the 95% overall uptime target could not be completely tested out as the power supply problem was solved by adding a higher rated buck converter and applying a staggered servo initialisation. Full workflow testing validated that the entire user journey, from QR code scanning to classification and reward crediting was successful through end-to-end operation, in practice.

Fifth, the results of the survey were a robust validation of the system design decisions given a context. About 76.4% of respondents also indicated this is a willingness to use VERIDIS, if used in a reliable form on their campus. Money on mobile was the most rewarding (58.9%), as 80.4% of them listed earning mobile money as the most influential of the motivations. The leaderboard utility resonated with 58.9% of participants and 55.4% self-reported engaging in improper disposal behaviour (always or sometimes) to demonstrate behaviour gap addressing role the system is aimed to fulfil.

In general, we have shown that a low-cost ESP32 reverse vending machine with sensor fusion can provide commercial classification accuracy (96%) with instant mobile money incentives fitting with the payment needs and preferences of Ghanaian university students. The results add to the body of literature on technology-based waste management in developing countries and a pathway for campus sustainability efforts in West Africa in a replication model.

5.2 Conclusion

This paper demonstrated the design, development, and partial validation of a smart incentive-based reverse vending system, VERIDIS, to promote responsible segregation of waste in Ghanaian university campuses. Its modular architecture, which integrated AI-based vision classification, light-based material sensing, IoT interface, and mobile app, was technically feasible on affordable and accessible hardware. The ESP32 microcontroller managed multiple I2C and SPI integrated nodes simultaneously by running a web server, accepting sensor responses, and producing a graphical TFT display, proving that a well-crafted embedded device can be developed within a constrained budget appropriate for the Ghanaian environment.

The majority voting classification scheme and the Flutter mobile app were the two most successful results of the project. Even though the ToF sensor was removed, the two-sensor system exhibited complementary advantages with HuskyLens for visual patterning and LDR for material transparency. The developed app was well-integrated with user authentication, QR-based session setting via deep linking, live bottle counting, simulated payment of mobile money rewards, transaction history, and live leaderboard, which all communicated with the ESP32 web server over a shared Wi-Fi network.

The project also had to deal with extensive hardware integration issues it faced towards the end of the project like the servo being non-responsive, ToF sensor connectivity failure, and touch

screen pin constraint. All these problems led to an incomplete fully automated end-to-end demonstration of the originally intended system. Not that they compromise the soundness of the entire system architecture, only underline the need to ensure adequate hardware debugging and component-level validation timeframes across future versions of the project.

The discovery of this study supports the conclusion that VERIDIS is a contextually appropriate and technically grounded approach to the solution in the area of campus waste segregation in Ghana. The survey results and available literature confirm that Ghanaian university students are willing to segregate waste when given the correct infrastructure and financial support for the system, and VERIDIS tackles them with an automatic classification and mobile money reward method, directly related to these concerns. Phase 2 development with resolved hardware and formal user testing verification would justify the implementation of VERIDIS if it were to proceed to a genuine campus deployment.

5.3 Limitations

VERIDIS suffers from the following limitations that were encountered during its development and testing:

- The HuskyLens AI camera runs on firmware 0.5.1b, necessitating a microSD card to save trained classification models permanently. All trained object IDs, such as plastic (ID 1), glass (ID 2), rejected items (ID 3), and empty chute (ID 4), are discarded on every power cycle without an SD card. This required maintaining continuous power to the HuskyLens throughout the demonstration, rendering it impractical for a real-world deployment and leading to significant setup time before each test run. For object classification, the HuskyLens utilizes a KNN (K-Nearest Neighbours) algorithm. Plastic and glass bottles are visually similar and there is no consistency with the appearance-based identification. Problems with IDs 1 and 2 were indeed still present despite long

training with more than 475 samples for each class, which is a basic limitation of vision-based material classification using KNN over visually similar objects.

- The VL53L0X ToF sensor was never confirmed as functioning at any point in the process. Despite an I2C bus that worked correctly for other devices including the HuskyLens at address 0x32 and PCA9685 at address 0x40, the ToF sensor never appeared at its expected address (0x29), despite extensive wiring and software testing. This removed one of the three planned classification sensors from the majority voting system, reducing it from a three-sensor to a two-sensor system comprising the HuskyLens and LDR only.
- After extensive testing and calibration sessions in which all six servo motors responded correctly to PWM signals from the PCA9685, the servos simultaneously stopped responding in the final stages of testing. Despite I2C communication between the ESP32 and PCA9685 remaining fully functional and software diagnostics confirming that commands were being sent correctly to all 16 channels, no servo movement could be achieved. This prevented full mechanical routing of bottles during the final demonstration and meant the system could not be tested end-to-end as originally designed.
- The XPT2046 touch controller shared the SPI bus with the TFT display. GPIO 35, which was assigned as the touch chip select pin, is an input-only pin on the ESP32 and cannot function as a chip select output. This prevented touch input from working entirely throughout the project and was not identified until late in the development process. As a result, the DONE button was removed from the TFT screen and relocated to the mobile application, representing an unplanned hardware constraint that altered the original system design.
- Material Scope Restricted: The system classifies only plastic and glass bottles. Other

recyclable materials common on campuses (aluminium cans, paper, cartons) are not supported, limiting the system's waste diversion potential.

- **Mobile Money Dependency:** The reward system requires users to have a registered mobile money number (MTN MoMo or equivalent), which may exclude some international students or staff without Ghanaian SIM cards.

5.4 Recommendations

The first and most important immediate improvement is obtaining a FAT32-formatted microSD card for the HuskyLens for permanent model storage across power cycles. In the future trainings, we are expecting to have a much larger and a much diverse dataset that will cover the entire spectrum of bottle brands, sizes, colours as well as conditions that will be collected at exactly the same lighting conditions inside the entry chute. Training a dedicated background and empty class is generally the best practice to avoid false detections when you have no bottle, and adding other samples with different lighting intensities could also result in a larger dataset, and would greatly enhance the robustness of the model and how consistently it can be classified.

The HuskyLens can also be replaced with an OpenMV Cam H7 or a Raspberry Pi Zero running a custom TensorFlow Lite model. They help to implement advanced classification algorithms, have bigger training instances, are flexible to discriminate materials by texture, surface reflectance and shape all at once. This would greatly increase the classification power of the same materials like plastic and glass bottles and we would close the hole of KNN algorithm that one can not successfully identify materials with the similar visual features.

Additionally, the VL53L0X ToF sensor wiring should be rechecked systematically,

including verifying the voltage level, test for swapped SDA and SCL wires, connect XSHUT to 3.3V, replace the module if required. If this hardware factor is resolved, the system will move substantially nearer to the original design specification and full E2E testing can be performed. Implementing MQTT over campus WiFi, or access point mode on ESP32, in production deployment, would allow MQTT to be a server isolated communication from infrastructure and a standalone protocol. Each physical machine must possess a unique machine specific QR code encodable with machine specific data code for the system to differentiate between the different deployed devices throughout campus and monitor bottle counts and bin fill each machine independently for monitoring dashboards for campus managers and structured sharing of data between recycling partners like SESA Recycling.

Exploring a near-infrared sensor (NIR) to this system, which will be mounted on the chute, since NIR spectroscopy provides full material identification by molecular composition instead of a visual image and would eliminate the key limitation of sight-based classification with materials that visually cannot be distinguished.

REFERENCES

- ABC Research. (2022). *Incentivization by blockchain technology: Reducing plastic waste through blockchain-based incentivization – The case of the plastic bottle supply chain* [FFG Success Story]. https://www.abc-research.at/wp-content/uploads/2022/03/FFG_Success-Story.pdf
- Adu-Boahen, K., Atampugre, G., Antwi, K. B., Osman, A., Osei, K. N., Mensah, E. A., & Adu-Boahen, A. O. (2014). Waste management practices in Ghana: Challenges and prospect, Jukwa Central Region. *International Journal of Development and Sustainability*, 3(3), 530-546.
- Adaji, I., Idoko, P., & Adisa, M. O. (2024, June). Insights from the review of apps that influence environmental sustainability. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization* (pp. 154-159).
- Agyei, A. (2020). AGYEI 2020 [Document on Scribd]. Scribd. <https://www.scribd.com/document/677915897/AGYEI-2020>
- Ahmad, G., Aleem, F. M., Alyas, T., Rizwan, A., & Raza, M. (2024). Intelligent waste sorting for urban sustainability using deep learning. *Scientific Reports*, 14(1), 27078. <https://doi.org/10.1038/s41598-024-78461-9>
- Akomea-Frimpong, I., Tetteh, P. A., Ofori, J. N. A., Tumpa, R. J., Pariafsai, F., Tenakwah, E. S., & Kukah, A. S. (2023). A review of barriers to circular economy implementation in solid waste management. *Journal of Cleaner Production*, 418, 138136.
- ALBA. (2025, January 6). Celebrating one year of success with the RVM project in Hong Kong. ALBA. <https://www.alba.com/news/celebrating-one-year-of-success-with-the-rvm-project-in-hong-kong/>

- Annas, M. A., Widodo, A., Aisiyah, M. C., Ningrum, I. E., & Makrufah, D. (2022). Karakterisasi sensor cahaya light-dependent resistor (LDR). *MASALIQ*, 2(4), 612–622. <https://doi.org/10.58578/masaliq.v2i4.516>
- Araujo-Andrade, C., Bugnicourt, E., Philippet, L., Rodriguez-Turienzo, L., Nettleton, D. F., Hoffmann, L., & Schlummer, M. (2021). Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes in view of their posterior recycling. *Waste Management & Research*, 39(5), 631–651. <https://doi.org/10.1177/0734242X209783>
- Argirusis, N., Achilleos, A., Alizadeh, N., Argirusis, C., & Sourkouni, G. (2025). IR sensors, related materials, and applications. *Sensors*, 25(3), 673. <https://doi.org/10.3390/s25030673>
- Baek, C. R., Kim, H. D., & Jang, Y.-C. (2025). Exploring glass recycling: Trends, technologies, and future trajectories. *Environmental Engineering Research*, 30(3), Article 240241. <https://doi.org/10.4491/eer.2024.241>
- BEKTAŞ, S. (n.d.). *The study of beverage container recycling process and potential market for Reverse Vending Machine (RVM) in Japan* [Unpublished manuscript].
- Bhalerao, S., & Ingle, M. (2021). Convolutional neural network: A systemic review and its application using Keras. In P. Karuppusamy, J. Perikos, I. G. B. B. Nugroho, & Y. Ma (Eds.), *Computing technologies and applications* (pp. 121-132). CRC Press. <https://doi.org/10.1201/9781003166702-11>
- Debrah, E. W. (2025, June 5). World Environment Day: Act now or choke later: Ghana's plastic time bomb. UESD. <https://www.uesd.edu.gh/world-environment-day-act-now-or-choke-later-ghanas-plastic-time-bomb/>

- Dery, F. A., Kuusaana, E. D., & Owusu-Sekyere, E. (2018). Solid waste characterization and recycling potential for university campuses in Ghana: Case study of two Ghanaian universities. *Journal of Waste Recycling*, 3(1), 3.
- Dugbazah, J., Glover, B., Mbuli, B., & Kungade, C. (2021, July 19). What a waste: Innovations in Africa's waste material management. AUDA-NEPAD.
<https://www.nepad.org/blog/what-waste-innovations-africas-waste-material-management>
- Durban University of Technology. (2024, September 26). DUT launches Reverse Vending Machine project to foster green minds and sustainable practices.
<https://www.dut.ac.za/dut-launches-reverse-vending-machine-project-to-foster-green-minds-and-sustainable-practices/>
- EcoATM. (2023). How ecoATM works. ecoATM. <https://www.ecoatm.com/how-it-works/>
- Elmenreich, W. (2002). *Sensor fusion in time-triggered systems* [Doctoral dissertation, Vienna University of Technology].
https://mobile.aau.at/~welmenre/papers/elmenreich_Dissertation_sensorFusionInTimeTriggeredSystems.pdf
- Florence Dery. (2017). *Investigating university students' knowledge and practices towards appropriate solid waste management practices in the Greater Accra Region, Ghana*. GRIN Verlag. <https://www.grin.com/document/358704>
- Goud, E. G., Manikanta, P., & Varshith, R. (2023). Smart waste management system using Arduino. *International Journal of Scientific Research in Engineering and Management (IJSREM)*, 7(3). <https://doi.org/10.55041/ijsrem18230>
- Jagannathan, U. K., Balaji, N., Brunda, A., & Gagan, M. B. (2022, March). QR code enabled domestic waste segregation tracking. In *Proceedings of the International Conference on Industrial Engineering and Operations Management* (pp. 7-10), Istanbul.

- Kaya, M., Ulutürk, S., Çetin Kaya, Y., & Altıntaş, O. (2023). Optimization of several deep CNN models for waste classification. *Sakarya University Journal of Computer and Information Sciences*, 6(2), 91–104.
<https://dergipark.org.tr/en/pub/saucis/issue/79575/1257100>
- Keep Scotland Beautiful, New College Lanarkshire, & Coca-Cola Europacific Partners GB. (2025). 20p incentive drives eight-fold surge in campus DRS pilot
- Kretschmer, M., Pennekamp, J., & Wehrle, K. (2021). Cookie banners and privacy policies: Measuring the impact of the GDPR on the web. *ACM Transactions on the Web (TWEB)*, 15(4), 1-42.
- Li, X., & Grammenos, R. (2023). Evaluation of practical edge computing CNN-based solutions for intelligent recycling bins. *IET Smart Cities*, 5(3), 194-209. <https://doi.org/10.1049/smc2.12057>
- Litoclean. (2024). The World Bank supports solid waste management in West Africa. Litoclean. <https://www.litoclean.com/en/blog/the-world-bank-supports-solid-waste-management-in-west-africa>
- Ministry of Environment, Science, Technology and Innovation (mesti). (2020). National solid waste management strategy for Ghana
- Mao, W., Chen, W., Wang, C., & Lin, Y. (2020). Recycling waste classification using optimized convolutional neural network. *Resources, Conservation and Recycling*, 164, 105132. <https://doi.org/10.1016/j.resconrec.2020.105132>
- Nasir, I., & Al-Talib, G. A. A. (2023). Waste classification using artificial intelligence techniques: Literature review. *Technium: Romanian Journal of Applied Sciences and Technology*, 5, 49–59. <https://doi.org/10.47577/technium.v5i.8345>
- National Plastic Action Partnership (NPAP). (2021). Ghana. Global Plastic Action Partnership. <https://www.globalplasticaction.org/ghana>

- Ruj, B., Pandey, V., Jash, P., & Srivastava, V. (2015). Sorting of plastic waste for effective recycling. *International Journal of Applied Science and Engineering Research*, 4, 564-571.
- Sackey, L. N., Anum, D. N. A., Ampiah, E. Y., Marfo, C. A., Amos, K., Ahorlley, N. A. A., Agyemang, L. S. Y., & Addy, H. S. T. (2025). Assessing the efficiency of bin-centered solid waste segregation system in Ashanti Region, Ghana. *The Scientific World Journal*, 2025, Article 6169623. <https://doi.org/10.1155/tswj/6169623>
- Saeed, A.-R., & Adodoadji-Dogbe, C. (2025, June 5). Ending plastic pollution for sustainability. UNDP Ghana. <https://www.undp.org/ghana/blog/ending-plastic-pollution-sustainability>
- Safo-Adu, G., & Owusu-Adzolah, N. (2023). Solid waste characterisation and recycling potential: A study in secondary schools in Kumasi Metropolis, Ghana. *Cleaner Waste Systems*, 4, 100065.
- Saha, S., Hemal, M. M., Abedin Eidmum, M. Z., & Mridha, M. F. (2025). Knowledge distillation approach for skin cancer classification on lightweight deep learning model. *Healthcare Technology Letters*, 12(1), e12120. <https://doi.org/10.1049/htl2.12120>
- Squire, J. N. T. (2016). Environmental and public health effects of plastic packaging debris in Accra, Ghana: Assessing a policy role for extended producer responsibility (EPR). *Journal of Geography, Environment and Earth Science International*, 6(2), 1– 6. <https://doi.org/10.9734/JGEESI/2016/25541>
- Statista Research Department. (2025, March 7). Plastic waste worldwide – statistics & facts. Statista. <https://www.statista.com/topics/5401/global-plastic-waste/>
- Sustainability Directory. (n.d.). Intrinsic environmental motivation. Lifestyle – Sustainability Directory. <https://lifestyle.sustainability-directory.com/term/intrinsic-environmental-motivation/>

- Tang, F.-L. (2011). *The study of beverage container recycling process and potential market for reverse vending machine (RVM) in Japan* (Master's research report). Ritsumeikan Asia Pacific University, Graduate School of Asia Pacific Studies.
- TOMRA. (n.d.). Deposit-refund system. TOMRA. <https://www.tomra.com/reverse-vending>
- TOMRA. (2022, April 11). Supermarkets can now offer digital recycling refunds through their store loyalty programs. TOMRA. <https://www.tomra.com/reverse-vending/media-center/news/2022/supermarkets-can-now-offer-digital-recycling-refunds-through-their-store-loyalty-programs>
- University of Glasgow. (2018, July 4). UofG installs 'reverse vending machine'. MyGlasgow. https://www.gla.ac.uk/myglasgow/news/peopleprojects/headline_596137_en.html
- Venturi, S., Zulauf, K., Cuel, R., & Wagner, R. (2025). Trash to treasure: Gamification and informed recycling behavior. *Resources, Conservation and Recycling*, 215, 108108.
- World Bank. (2022, February 11). Solid waste management [Brief]. <https://www.worldbank.org/en/topic/urbandevelopment/brief/solid-waste-management>
- Yaxi, X., & Yuling, Z. (2020). Object detection using deep learning and camera-lidar fusion methods.
- Yang, X., & Thøgersen, J. (2022). When people are green and greedy: A new perspective of recycling rewards and crowding-out in Germany, the USA, and China. *Journal of Business Research*, 144, 217-235. <https://doi.org/10.1016/j.jbusres.2022.01.086>

APPENDIX A: GITHUB REPOSITORY

The complete source code for the VERIDIS mobile application, including all Flutter screens, Firebase integration, session management, leaderboard, and reward logic, is available at the following GitHub repository:

https://github.com/GyameraYaw/VERIDIS_app

APPENDIX B: SURVEY INSTRUMENT

Survey Link

The complete survey used to collect data for VERIDIS is available at the following link:

https://docs.google.com/forms/d/e/1FAIpQLSeRRMzgbWPW__O4veiQ9BZtAZ-GgmjtMu1TsPsvQ80l13g4hw/viewform?usp=sharing&ouid=106244432392375410090

Survey Questions

1. Do you consent to take part in this survey?
2. What is your gender?
3. What is your age range?
4. What is your occupation or role?
5. Which institution are you in?
6. Have you ever participated in a recycling activity before?
7. If yes, what type of recyclables did you recycle or collect?
8. How important do you think recycling is for the environment?
9. How often do you separate your waste (waste segregation)?
10. How often do you currently recycle plastic or glass bottles?
11. What are the main reasons people don't recycle or separate waste regularly (in your opinion)?
12. Would you be interested in using a smart recycling station that rewards you for recycling bottles?
13. Which of these features would make the VERIDIS more appealing to you?
14. How easy would you expect it to be to use a system that identifies bottle type automatically (plastic vs glass)?
15. Would you trust an AI and sensor-based system (IR sensor + LDR sensor + camera) to

correctly identify and sort bottles?

16. What additional features would make you trust and use the system more?
17. How important is it for the system to display your real-time progress (number of bottles deposited, amount earned, etc.)?
18. Would you like the system to allow you to view your waste history through the mobile app (history of glass and plastic bottles segregated)?
19. What would motivate you more to recycle using this system?
20. What would you prefer to receive rewards in?
21. How familiar are you with blockchain or cryptocurrency systems (e.g., Polygon, Binance, Ethereum)?
22. Would you be willing to connect a digital wallet (e.g., MetaMask, Trust Wallet) to receive recycling rewards?
23. Do you think introducing such a system on university campuses will increase students' participation in recycling?
24. How likely are you to encourage others to use this system if implemented?
25. What impact do you think rewards will have on recycling habits among the youth?
26. Would you support a future expansion of this system to other waste types (e.g., cans, paper, e-waste)?
27. Open Feedback?
28. What do you like most about the idea of this waste station?
29. What concerns or challenges do you foresee with this system?
30. What suggestions would you make to improve the system before it's fully developed?